



RESEARCH INTERNSHIP REPORT

Learning Climate-Invariant Representations for Out-of-Distribution Climate Generalization

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Abstract

Machine learning models are increasingly used in climate science for prediction, emulation, and physical parameterization tasks. However, their deployment under climate change raises a fundamental challenge: models are typically trained on historical climate data but are expected to generalize to future climate conditions whose statistical properties may differ substantially from those observed during training. This distribution shift can lead to significant degradation in predictive performance and limits the reliability of machine learning methods for climate applications.

This work investigates whether learning climate-invariant latent representations can improve generalization across climate states. In particular, we study the Climate-Equivariant Representation Alignment (CERA) framework, which aims to align latent representations across climates while preserving information relevant to downstream prediction tasks. We evaluate several architectures, including supervised baselines, autoencoders, aligned autoencoders, and a CERA-inspired model, using CMIP6 climate simulations spanning historical conditions and multiple future Shared Socioeconomic Pathway (SSP) scenarios.

Experiments are conducted on prediction tasks of varying difficulty, including temperature, humidity, precipitation, and masked-point reconstruction. Beyond predictive performance, we analyze the geometry of the learned latent spaces through Wasserstein distances, and low-dimensional projections in order to quantify climate alignment and investigate its relationship with out-of-distribution generalization.

A particular focus is placed on the physical interpretation of latent representations. To determine whether CERA learns meaningful climate-invariant structures rather than merely enforcing latent alignment, we analyze a range of physically motivated quantities, including relative humidity, atmospheric stability indicators, wind-related variables, and climate anomalies. We further investigate latent neighborhoods across climates and explore data-driven approaches for identifying emergent invariants through sparse and symbolic regression. These analyses aim to assess whether latent similarity corresponds to physically meaningful similarity and whether the learned representations capture robust physical structures that persist across climate conditions.

Finally, we discuss the implications of climate-invariant representation learning for climate foundation models such as ClimaX. While large pretrained models provide powerful climate representations, our results suggest that explicit mechanisms for climate invariance may remain necessary to achieve robust out-of-distribution generalization under climate change.

Overall, this work contributes to a better understanding of the relationship between latent alignment, physical invariance, and climate generalization, and provides a framework for evaluating whether learned climate representations capture meaningful and transferable physical information.

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1 Introduction

Recent years have witnessed remarkable progress in the application of machine learning to the Earth sciences. Deep learning models are now routinely used for weather forecasting, climate emulation, statistical downscaling, data assimilation, and the representation of unresolved physical processes. More recently, the emergence of climate foundation models has demonstrated that large-scale pretrained architectures can learn representations transferable across multiple climate-related tasks. These developments have generated considerable interest within the climate science community, where machine learning is increasingly viewed as a key tool for extracting information from large climate simulations and accelerating computationally expensive modeling workflows.

However, applying machine learning to climate science presents a fundamental challenge that is less pronounced in many traditional machine learning applications: the climate system is continuously evolving. Unlike image classification or language modeling tasks, where the underlying data-generating process is often assumed to remain approximately stationary, climate data are inherently non-stationary. Anthropogenic greenhouse gas emissions modify the Earth’s radiative balance and induce long-term changes in temperature, humidity, atmospheric circulation, precipitation patterns, and many other variables. Consequently, the statistical distributions characterizing future climates may differ substantially from those observed during the historical period.

This evolution of the climate system creates a distribution shift problem. Most machine learning models are trained using historical observations or historical climate simulations because these datasets are generally more abundant, better constrained, and more reliable than projections of future climate conditions. Yet the ultimate objective of many climate applications is precisely to make predictions under future climates. Models trained on present-day conditions must therefore be deployed in environments that may exhibit statistical properties never encountered during training.

This mismatch between training and deployment conditions lies at the heart of the broader problem of out-of-distribution (OOD) generalization. Although modern machine learning models often achieve excellent performance when evaluated on data drawn from the same distribution as the training set, they are well known to degrade when confronted with distribution shifts. In the climate context, such failures can have important consequences. A model may accurately predict a target variable under historical conditions while producing increasingly unreliable predictions as the climate warms. Understanding how to construct models that remain robust under changing climatic conditions therefore constitutes a major scientific challenge at the intersection of machine learning and climate science.

A growing body of research suggests that robust OOD generalization requires learning representations that capture fundamental properties of the underlying physical system rather than climate-specific statistical correlations. In climate science, this idea motivates the search for climate-invariant representations: latent variables that remain meaningful across different climate states while preserving information relevant to downstream prediction tasks.

One particularly promising approach in this direction is the Climate-Equivariant Representation Alignment (CERA) framework. CERA introduces an explicit alignment mechanism that encourages latent representations originating from different climates to occupy similar regions of the latent space. The underlying hypothesis is that a suitable latent representation can separate climate-invariant information from climate-dependent variability, thereby improving transferability across climate conditions. While encouraging results have been reported in specific climate modeling applications, several fundamental questions remain open. In particular, it is still unclear whether latent alignment genuinely improves climate generalization, whether the resulting

representations correspond to physically meaningful invariants, and to what extent similar structures are already learned by modern climate foundation models.

The objective of this internship is to investigate the relationship between latent representation alignment, physical invariance, and out-of-distribution climate generalization. More specifically, this work aims to answer three main scientific questions. First, does explicit climate alignment improve predictive performance when models are evaluated on climates warmer than those encountered during training ? Second, do the latent representations learned by CERA encode physically meaningful climate-invariant structures, and to what extent does explicit latent alignment contribute to the emergence of these invariants ? Third, how does explicit representation alignment compare with the representations learned by large pretrained climate foundation models such as ClimaX ?

To address these questions, we evaluate several architectures ranging from standard supervised baselines and conventional autoencoders to aligned autoencoders and a CERA-inspired architecture. The experiments are conducted using CMIP6 climate simulations covering historical conditions as well as multiple future Shared Socioeconomic Pathway (SSP) scenarios. We consider prediction tasks of varying difficulty, including temperature, precipitation, and masked-point reconstruction, in order to assess both predictive performance and robustness under climate change.

Beyond predictive performance, a central aspect of this work is the analysis of the learned latent spaces. We investigate whether explicit climate alignment promotes the emergence of climate-invariant representations and whether these representations capture structures that remain stable across climate conditions. We then examine the physical meaning of these invariants by assessing their relationship to established climate variables and physically motivated quantities, with the goal of determining whether the learned latent organization reflects genuine physical invariants rather than merely statistical similarities.

The contributions of this work are fourfold. First, we assess the extent to which the benefits of CERA and climate representation alignment translate to realistic CMIP6 climate simulations, providing a validation of the framework beyond its original experimental setting. Second, we investigate the interplay between latent alignment, climate-invariant representation learning, and out-of-distribution climate generalization across several prediction tasks. Third, we introduce a framework for assessing whether learned climate representations encode physically meaningful invariants by combining latent-space analysis, cross-climate neighborhood comparisons, and invariant discovery methods. Fourth, we discuss the implications of these findings for climate foundation models and examine whether explicit invariance mechanisms may complement large-scale pretraining strategies.

The remainder of this report is organized as follows. Section 2 reviews related work, Section 3 presents the datasets and experimental setup, Section 4 describes the proposed methods, Sections 5 and 6 present the experimental results and latent-space analyses, and Section 7 concludes the report.

2 Related Work

2.1 Climate Models and the Generalization Problem

Recent advances in machine learning have profoundly transformed weather and climate modeling. Traditional Earth System Models (ESMs) remain the primary tool for climate projection, yet their computational cost limits the number of simulations that can be produced and analyzed. As a consequence, a growing body of research has explored data-driven alternatives capable of learning climate dynamics directly from large collections of observations and numerical simulations.

In particular, the recent success of deep learning has led to the emergence of foundation models in weather and climate science. Foundation models are large machine learning systems trained on broad and diverse datasets, typically through self-supervised learning, and designed to be adapted to a wide range of downstream tasks rather than a single prediction problem [1]. Originally introduced in natural language processing and computer vision, this paradigm has recently been extended to Earth system sciences, where large-scale pretraining is used to learn transferable representations from heterogeneous geophysical datasets.

Several climate foundation models have been proposed in recent years. ClimaX [2] was among the first transformer-based foundation models designed specifically for weather and climate applications, while Aurora [3] recently extended this approach to a broader range of Earth system forecasting tasks. These models are typically pretrained on large collections of historical observations, atmospheric reanalyses such as ERA5, and climate simulations from CMIP6.

The reliance on historical data is a fundamental characteristic of climate machine learning. Training modern deep learning models requires massive amounts of data, and the most abundant and reliable datasets currently available correspond to historical climate conditions. Consequently, most machine learning models are primarily exposed during training to the statistical properties of past climates.

However, the climate system is not stationary. Anthropogenic greenhouse gas emissions continuously modify the Earth’s radiative balance, resulting in long-term changes in temperature, humidity, atmospheric circulation, precipitation patterns, and extreme events. According to the IPCC Sixth Assessment Report, anthropogenic climate change is already modifying the frequency, intensity, and statistical characteristics of many climatic variables and extreme events, with further changes projected under continued warming [4]. As a consequence, future climates are characterized by probability distributions that increasingly diverge from those observed historically.

From a machine learning perspective, this phenomenon naturally gives rise to a distribution shift problem. Models trained on historical climate conditions are ultimately expected to make predictions in future climates whose statistical properties differ from those encountered during training. This setting falls within the broader framework of out-of-distribution (OOD) generalization. As discussed by Liu et al. [5], OOD generalization refers to the ability of a model to maintain predictive performance when deployed on data drawn from a distribution that differs from the training distribution. Such shifts are particularly challenging because machine learning models often exploit statistical correlations that may not remain valid outside the environments in which they were learned.

Climate change provides a particularly severe instance of OOD generalization. Unlike conventional machine learning benchmarks where distribution shifts often arise from changes in acquisition conditions or domain-specific biases, climate shifts are driven by evolving physical forcings acting on a nonlinear dynamical system. Future climates therefore represent progressively changing environments rather than independent domains.

Several studies have highlighted the impact of climate distribution shifts on machine learning models. In a seminal work, O’Gorman and Dwyer [6] investigated machine-learning-based parameterizations of moist convection and evaluated their ability to generalize across different climate states. As illustrated in Figure 1, models trained exclusively on a control climate struggled to reproduce the response of the system under warmer conditions, whereas training on a broader range of climates substantially improved generalization.

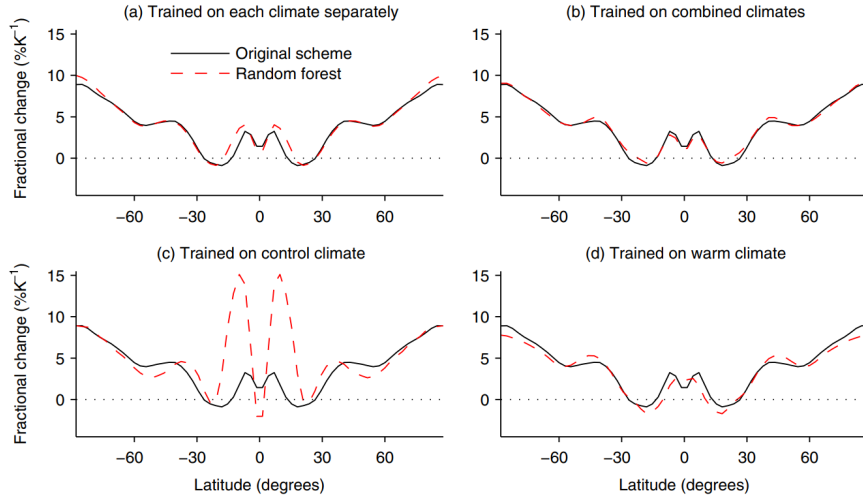


Figure 1: Impact of the training climate on the ability of a machine-learning parameterization to reproduce precipitation changes under warming. Models trained on multiple climate states generalize more successfully across climates than models trained on a single climate [6]

These results provide direct evidence that climate distribution shifts can limit the ability of machine learning models to extrapolate beyond the climatic conditions represented in their training data.

One possible strategy for mitigating this problem is to increase the diversity of the training distribution by leveraging large-scale pretraining and incorporating climate simulations spanning both historical and future scenarios. By exposing the model to a broader range of climate states during training, the goal is to extend the support of the training distribution and improve robustness to future climate conditions. This is the philosophy underlying climate foundation models such as ClimaX [2], which are trained on heterogeneous climate datasets spanning multiple variables, spatial resolutions, simulation sources, and climate regimes. The underlying hypothesis is that exposure to a broader set of climate conditions may enable the model to learn more transferable representations.

Nevertheless, recent evidence suggests that large-scale pretraining alone does not fully resolve the challenge of climate extrapolation. Navarro et al. [7] evaluated several climate emulators under controlled no-analog distribution shifts by restricting downstream training to historical climate conditions (1850–2014) and evaluating performance on future forcing scenarios. Importantly, while ClimaX relies on a backbone pretrained on a large collection of heterogeneous CMIP6 climate simulations, including future climates, it was fine-tuned exclusively on historical data, providing a controlled test of its ability to generalize beyond the historical climate distribution.

As shown in Figure 2, ClimaX consistently achieves the lowest absolute prediction errors across most tasks, which seems to confirm the benefits of large-scale pretraining. However, it also exhibits a larger relative degradation as the climate shift increases, particularly under the SSP5-8.5 scenario.

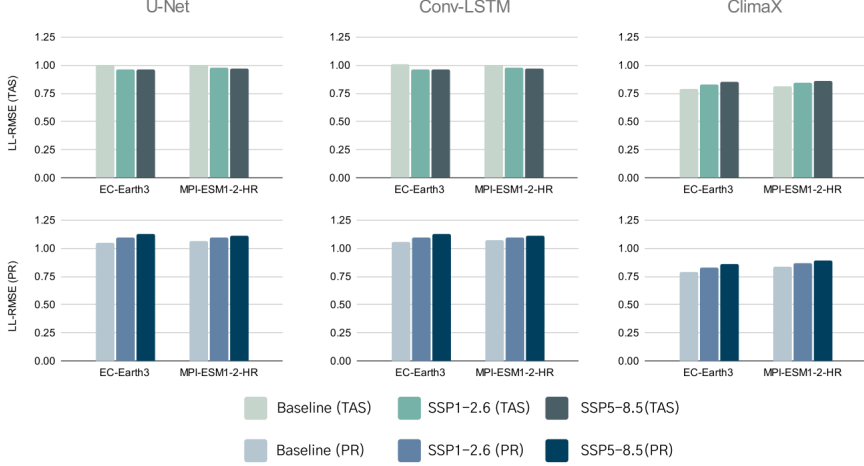


Figure 2: Performance of U-Net, ConvLSTM, and ClimaX under increasing climate distribution shifts. While ClimaX achieves the best absolute performance, it exhibits a larger relative degradation under future climate scenarios, particularly for precipitation prediction. [7]

This effect is more pronounced for precipitation prediction. Although temperature errors also increase under stronger forcing scenarios for ClimaX, the degradation remains larger for precipitation as the climate departs from historical conditions (Figure 2). This behavior likely reflects the highly nonlinear response of precipitation to climate change, which depends on complex interactions between thermodynamics, atmospheric circulation, convection, and regional feedbacks.

Navarro et al. [7] discuss several hypotheses that may explain this stronger sensitivity to climate distribution shifts. Despite being pretrained on a diverse collection of climate simulations, ClimaX may remain partially anchored to statistical patterns learned during pretraining rather than to physical mechanisms that generalize across forcing regimes. The authors further suggest that the global attention mechanisms used by Transformer architectures could be more sensitive to changes in forcing magnitude than the local spatial priors of convolutional models. While these explanations remain speculative, they provide plausible interpretations for the stronger relative degradation observed under severe climate shifts.

These findings suggest that increasing model scale and training data diversity is not, by itself, sufficient to guarantee robust generalization under climate change. Even foundation models exposed to future climate simulations during pretraining remain vulnerable to shifts in the underlying climate distribution. This observation raises a fundamental question: rather than relying on large-scale pretraining to implicitly discover transferable representations, can we explicitly encourage models to learn features that remain stable across climate conditions? Recent work has increasingly explored this direction through climate-invariant learning approaches, which aim to identify representations that are less dependent on climate-specific statistical patterns and more closely tied to the underlying physical processes governing the climate system. Such methods are reviewed in the next section.

2.2 Climate Invariant Learning

The difficulties encountered by machine learning models under climate distribution shifts naturally raise the question of what type of information should be learned in order to generalize beyond the climate conditions represented in the training data. A growing body of work suggests that successful extrapolation requires models to rely on relationships that remain valid across climates rather than on climate-specific statistical correlations.

This idea forms the basis of climate-invariant machine learning, introduced by Beucler et al. [8]. The central observation is that many atmospheric variables exhibit substantial distribution shifts as the climate changes. Temperature, specific humidity, precipitation, and radiative fluxes all evolve under warming, causing the statistical relationships learned from one climate to become unreliable in another. Rather than learning directly from these raw variables, Beucler et al. argue that machine learning models should be encouraged to operate on physically meaningful quantities whose distributions remain more stable across climate states.

The proposed framework consists of transforming the original climate variables into physically informed representations designed to be approximately invariant under climate change. Instead of predicting atmospheric processes from raw temperature and humidity fields, the model is provided with quantities derived from established physical principles. Examples include relative humidity rather than specific humidity, plume buoyancy rather than temperature, and normalized surface fluxes derived from bulk aerodynamic relationships. These transformations are motivated by known physical laws and are explicitly designed to reduce the distributional differences between cold and warm climates.

The underlying hypothesis is that machine learning models should generalize more effectively when they learn mappings expressed in terms of physically meaningful variables whose statistical properties remain approximately constant across climates. Beucler et al. demonstrate that this approach substantially improves the ability of neural-network parameterizations to generalize from colder to warmer climates across several atmospheric models. Their results show that physically transformed inputs lead to significantly lower out-of-distribution errors than models trained on raw climate variables.

Despite its promising results, this framework also presents important limitations. The invariant quantities must be identified manually by domain experts and explicitly engineered before training. Consequently, the approach is restricted to physical invariants that are already known and understood. The quality of the learned representation therefore depends heavily on prior scientific knowledge, and potentially important invariants may remain undiscovered.

These limitations motivate a complementary line of research that seeks to learn climate-invariant representations directly from data. Rather than manually specifying invariant physical variables, the objective is to discover latent representations whose organization remains stable across climates while preserving the information necessary for downstream prediction tasks. Representation alignment methods constitute one such approach and form the basis of the Climate-Equivariant Representation Alignment (CERA) framework discussed in the next section.

2.3 CERA

CERA is a framework proposed by Liu and O’Gorman [9], it seeks to learn climate-invariant representations directly from data through latent-space alignment. The central idea of CERA is illustrated in Figure 3. The framework consists of two jointly trained components: an autoencoder responsible for learning a compressed latent representation of atmospheric profiles, and a predictor trained to estimate the target climate variables from this latent representation. During training, samples originating from both a control climate and a warmer climate are passed through a shared encoder. The latent representations of the two climates are then explicitly aligned using an Earth Mover’s Distance (EMD) regularization term, while the predictor is trained exclusively using labels from the control climate. The objective is to encourage the encoder to extract information that is common to both climates while discarding climate-specific variability.

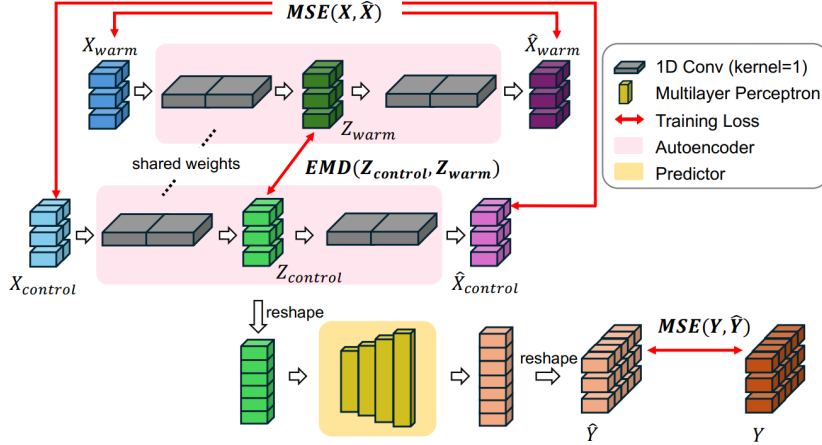


Figure 3: Overview of the Climate-Equivariant Representation Alignment (CERA) framework. Latent representations from a control and a warm climate are aligned through an Earth Mover’s Distance regularization, while prediction is performed using the aligned component of the latent space. [9]

The latent representation is intentionally partitioned into two components. One component is aligned across climates and used by the predictor, while the remaining dimensions are only used for reconstruction. The model is trained jointly through a composite objective combining prediction, reconstruction, and latent-alignment terms:

$$L = (1 - \lambda_{\text{rec}} - \lambda_{\text{align}}) L_{\text{pred}} + \lambda_{\text{rec}} L_{\text{rec}} + \lambda_{\text{align}} L_{\text{align}}, \quad (1)$$

where L_{pred} denotes the prediction loss on the control climate, L_{rec} is the autoencoder reconstruction loss, and L_{align} corresponds to the Earth Mover’s Distance (EMD) computed between the aligned latent distributions of the control and warm climates. The hyperparameters λ_{rec} and λ_{align} control the relative importance of reconstruction and alignment with respect to the prediction objective. This formulation encourages the predictor to rely on information that remains stable across climate conditions while allowing climate-dependent variability to be retained in the reconstruction pathway.

CERA was evaluated on the parameterization of moist atmospheric processes using idealized climate simulations generated with the System for Atmospheric Modeling (SAM). The experimental setup considers an aquaplanet configuration in which sea surface temperatures are uniformly increased by 4 K relative to a control climate. Atmospheric profiles of temperature and moisture are used as inputs, while various moisture and energy tendencies constitute the prediction targets. Importantly, the model has access to input data from both climates during training, but receives labels only for the control climate. The warmer climate therefore contributes exclusively through the latent alignment objective.

Within this controlled setting, CERA demonstrates promising results. The authors compare their approach against three baselines: a standard predictor trained on raw atmospheric variables, a physically informed baseline based on manually designed climate-invariant variables (relative humidity and plume buoyancy), and an ablation version of CERA without latent alignment. Across multiple moisture and energy prediction tasks, CERA consistently achieves the highest performance in the warmer climate while maintaining strong performance in the control climate. The ablation study further shows that latent alignment contributes significantly to the observed improvements, indicating that the alignment objective plays a central role in the model’s ability to generalize across climates.

These results provide an important proof of concept for data-driven climate-invariant representation learning.

Despite these promising results, several important questions remain open. First, CERA is evaluated in a highly idealized setting based on aquaplanet simulations and a single warming perturbation. While this framework provides a clean testbed for studying climate generalization, it remains unclear whether the same conclusions hold for more realistic Earth-system simulations involving multiple forcing scenarios, regional climate responses, and a larger diversity of climate regimes.

More fundamentally, the study provides limited insight into the nature of the learned representations themselves. The success of CERA is primarily assessed through downstream predictive performance, while the latent spaces receive relatively little analysis. As a result, several key assumptions remain largely untested: whether climate-invariant structures are actually learned, whether latent alignment is the mechanism responsible for their emergence, and whether these representations correspond to physically meaningful invariants. Similarly, the choice of Earth Mover’s Distance as an alignment metric, the use of a shared encoder, and the restriction of the predictor to the aligned latent dimensions introduce additional assumptions whose respective contributions are not explicitly disentangled. The authors themselves identify the interpretation of the learned latent representations as an important direction for future work. More generally, understanding how latent-space organization relates to climate invariance and out-of-distribution generalization remains an open research question.

These limitations naturally motivate a closer examination of representation learning and latent-space analysis methods. The next section therefore reviews some of the tools used to characterize, interpret, and compare learned representations.

2.4 Interpreting Learned Representations

The success of modern machine learning models is often attributed not only to their predictive capabilities, but also to their ability to learn internal representations that capture meaningful structure in the data. As emphasized by Bengio et al. [10], representation learning aims to transform raw observations into latent representations that reveal the underlying factors governing the observed phenomena. Consequently, understanding what information is encoded in a learned representation has become a central topic in modern machine learning research.

A large body of work has focused on the geometric analysis of learned representations. Visualization techniques such as PCA and t-SNE are commonly used to explore latent-space organization and reveal structures that may not be apparent in the original feature space. In particular, t-SNE has become a standard tool for visualizing high-dimensional representations [11]. These qualitative analyses are often complemented by quantitative measures of latent-space geometry, such as intrinsic dimensionality and effective rank, which can provide insight into the complexity and organization of learned representations [12]. And metrics such as Wasserstein distances or centroid distances are also frequently used to compare latent distributions across domains.

Beyond global geometric properties, another important question concerns the information encoded within a representation. A common strategy consists of using probing methods to determine whether specific attributes can be recovered from the latent space. Linear probes, nonlinear predictors, and classification tasks are frequently employed to assess the presence of domain, task, or physical information in learned representations. In particular, domain classifiers are often used to evaluate invariance: if the climate scenario associated with a sample remains easily identifiable from the latent representation, climate-specific information is still present, whereas poor domain separability may indicate a more invariant representation [13].

A complementary line of work focuses on interpreting the structure of the latent space itself. Clustering and neighborhood analyses are commonly used to identify coherent regions of the representation and determine whether nearby samples share common physical properties. More

recently, sparse regression and symbolic regression have been employed to uncover interpretable relationships and compact analytical descriptions of complex systems [14].

This perspective is particularly relevant for climate-invariant learning. While methods such as CERA aim to learn representations that generalize across climates, predictive performance alone does not reveal the nature of the structures encoded in the latent space. Assessing whether learned representations capture meaningful climate-invariant information therefore requires dedicated tools for analyzing their geometry, organization, and interpretability. The representation-learning literature reviewed above provides a conceptual framework for conducting such analyses.

Building on these previous studies, this work evaluates climate representation alignment on realistic CMIP6 simulations and investigates the relationship between latent alignment, climate invariance, and out-of-distribution generalization. The following sections present the datasets, experimental framework, and methodologies used to investigate these questions.

3 Data

3.1 CMIP6 Simulations

The experiments conducted in this study rely on climate simulations from the Coupled Model Intercomparison Project Phase 6 (CMIP6), which currently constitutes the reference international framework for the evaluation and intercomparison of Earth System Models (ESMs). CMIP6 coordinates a large ensemble of standardized climate simulations produced by research institutions worldwide, covering historical climate conditions as well as a range of future forcing scenarios designed to assess potential climate trajectories under different socioeconomic pathways [15].

A central objective of CMIP6 is to improve the understanding of how the Earth system responds to external forcings and to provide a common benchmark for studying past, present, and future climate change. The availability of simulations spanning multiple climate states makes CMIP6 particularly well suited for investigating machine-learning generalization under climate change, as it enables models to be trained and evaluated across progressively warmer climate conditions.

In this work, all climate variables are extracted from simulations generated with the Community Earth System Model version 2 (CESM2), one of the state-of-the-art Earth System Models participating in CMIP6 [16]. CESM2 provides physically consistent simulations of the coupled atmosphere, ocean, land, and sea-ice systems and has been widely used in climate-change research.

3.2 Climate Variables and Preprocessing

The dataset is constructed from multivariate atmospheric fields extracted from CESM2 climate simulations. The selected variables were chosen to provide a compact description of the thermodynamic and dynamical state of the atmosphere while retaining information relevant to temperature, moisture, circulation, and precipitation processes. The complete dataset contains sixteen climate variables. Depending on the prediction task considered, one variable is designated as the prediction target and removed from the model inputs, leaving fifteen input variables available to the learning algorithms.

Figure 4 summarizes the climate variables used throughout this study :

Variable	tas	ta850	ta500	huss	hus850	hus500
Description	Near-Surface Air Temperature	Air Temperature at 850 hPa	Air Temperature at 500 hPa	Near-Surface Specific Humidity	Specific Humidity at 850 hPa	Specific Humidity at 500 hPa
Variable	ua850	ua500	va850	va500	wap500	zg500
Description	Zonal Wind at 850 hPa	Zonal Wind at 500 hPa	Meridional Wind at 850 hPa	Meridional Wind at 500 hPa	Vertical Velocity at 500 hPa	Geopotential Height at 500 hPa
Variable	psl	rsds	sfcWind	pr		
Description	Sea Level Pressure	Surface Downwelling Shortwave Radiation	Near-Surface Wind Speed	Precipitation		

Figure 4: Climate variables used in this study. Depending on the prediction task, one variable is used as the prediction target while the remaining variables are provided as model inputs.

The climate fields are sampled at a daily temporal resolution and organized into multivariate spatial patches. Each sample corresponds to a geographical region of size 7×10 grid points, covering approximately $1000 \text{ km} \times 1000 \text{ km}$ and resulting in 70 spatial locations for each variable. The patches are extracted within the tropical belt, between 30°S and 30°N , yielding a total

of 246 fixed regions distributed around the globe. This sampling strategy captures a broad range of tropical climate regimes while focusing on regions where convection and precipitation play a central role in the climate system. For a given prediction task, each sample therefore consists of fifteen input variables observed over 70 grid points on a given day, together with the corresponding target variable. The complete dataset contains approximately 2.5 million samples for each climate scenario considered in this study.

Using spatial patches rather than individual grid points allows the models to exploit local spatial dependencies and interactions between neighboring atmospheric states. In addition, several variables are available at multiple pressure levels (surface, 850 hPa, and 500 hPa), providing a simplified representation of the vertical atmospheric structure. This vertical information is particularly important for processes related to moisture transport, atmospheric stability, and precipitation formation.

The dataset is divided into training, validation, and test sets using non-overlapping temporal periods. To avoid information leakage, all preprocessing operations are performed using statistics computed exclusively from the Historical training data. The resulting normalization parameters are then applied unchanged to all validation, test, and future-climate datasets.

For a given variable x , standardization is performed according to

$$x_{\text{norm}} = \frac{x - \mu_{\text{hist}}}{\sigma_{\text{hist}}},$$

where μ_{hist} and σ_{hist} denote the mean and standard deviation estimated on the Historical training set. Normalization is applied independently to each variable. Using a single normalization reference preserves the distribution shifts between climates while preventing information from future climates from influencing model training.

Precipitation requires a dedicated preprocessing step due to its highly skewed distribution, characterized by a large number of weak events and a small number of extreme values. Following common practice, precipitation is first converted from $\text{kg m}^{-2} \text{s}^{-1}$ to mm day^{-1} and transformed according to

$$pr_{\log} = \log(1 + pr_{\text{mm/day}}).$$

The transformed variable is subsequently standardized using the Historical training statistics. This logarithmic transformation reduces the dynamic range of precipitation values and improves the numerical stability of the learning process.

Three prediction tasks are considered throughout this study. The first task consists of predicting near-surface air temperature (**tas**), a relatively smooth variable whose response to climate change is comparatively well understood. The second task focuses on precipitation (**pr**), which represents a substantially more challenging target due to its intermittency and strong nonlinear dependence on atmospheric thermodynamics and circulation patterns. Finally, we consider a masked reconstruction task in which part of the input patch is intentionally hidden and must be reconstructed from the remaining information. By controlling the fraction and spatial organization of the masked regions, this task provides a flexible way to adjust the difficulty of the prediction problem and to evaluate the quality of the learned representations. Together, these tasks span different levels of complexity and provide complementary perspectives on climate generalization.

3.3 Climate Scenarios and Distribution Shift

Future climate simulations are drawn from the Scenario Model Intercomparison Project (ScenarioMIP), the CMIP6 framework dedicated to coordinated projections under alternative socioeconomic and greenhouse-gas emission pathways [17]. ScenarioMIP defines a set of Shared

Socioeconomic Pathway (SSP) scenarios spanning a wide range of future radiative forcing trajectories, from strong mitigation to high-emission futures. In this study, we consider four scenarios: SSP1-2.6, SSP2-4.5, SSP3-7.0, and SSP5-8.5. Together, these scenarios cover a broad spectrum of future climate conditions and warming levels, ranging from low-emission pathways to extreme warming trajectories. Figure 5 illustrates the evolution of global warming under the different SSP scenarios.

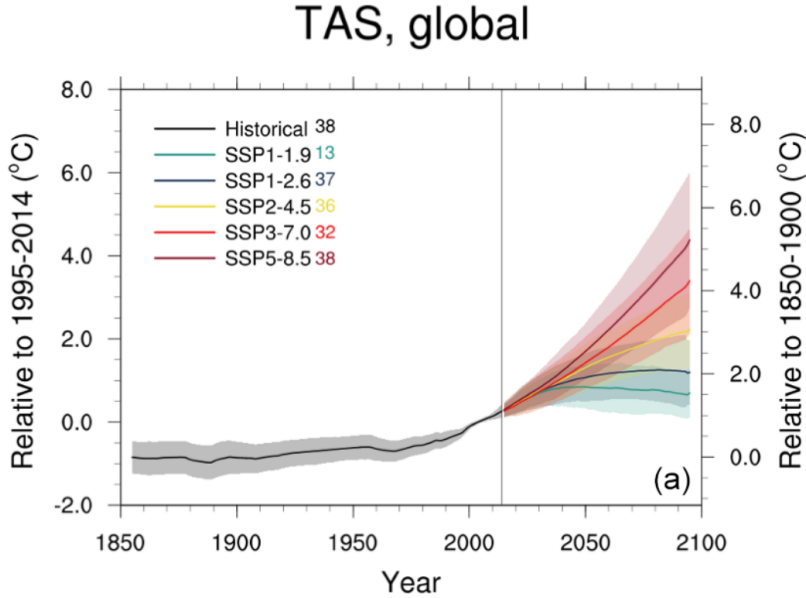


Figure 5: Evolution of global mean surface air temperature anomalies under the CMIP6 SSP scenarios. The scenarios span a wide range of future warming trajectories, from strong mitigation (SSP1-2.6) to high-emission futures (SSP5-8.5). [17]

The Historical climate (1850–2014) constitutes the reference environment and provides the labeled samples used for supervised training. SSP2-4.5 is used as an auxiliary climate during representation alignment, where samples are available without access to the target labels. The warmer SSP3-7.0 and SSP5-8.5 scenarios are subsequently used to evaluate the ability of the models to generalize under increasingly severe climate shifts. Finally, SSP1-2.6 provides an additional evaluation setting characterized by a strong mitigation pathway. Unlike the other scenarios, its forcing trajectory partially stabilizes during the second half of the century, resulting in a markedly different climate evolution and offering a complementary test of model robustness.

A central objective of this work is to study generalization under climate-induced distribution shifts. Before introducing the learning methods, it is therefore important to verify that such shifts are effectively present in the data. To this end, we analyze the distribution of multivariate climate patches across the different scenarios using Principal Component Analysis (PCA). The PCA is fitted on 100 000 randomly sampled patches from each climate, considering the complete set of climate variables available in each sample. Figure 6 shows the resulting distributions projected onto the first principal component (PC1).

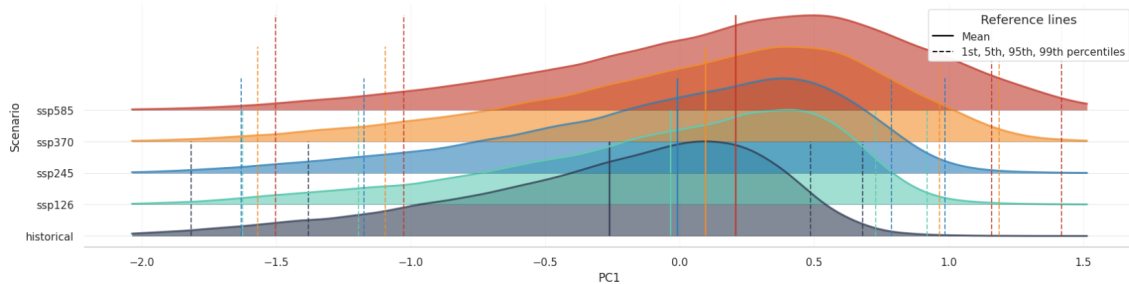


Figure 6: Distribution of multivariate climate patches projected onto the first principal component (PC1) for the Historical, SSP1-2.6, SSP2-4.5, SSP3-7.0 and SSP5-8.5 climates.

Several important trends emerge from figure 6. First, the center of the distributions progressively shifts toward larger PC1 values as radiative forcing increases. This indicates that the average state of the climate patches evolves substantially between the historical climate and future warming scenarios. Second, the distributions become increasingly spread out for warmer climates, with SSP3-7.0 and SSP5-8.5 exhibiting a noticeably larger variance than the Historical climate. This suggests that climate change not only modifies the mean state of the system but also expands the range of atmospheric conditions encountered in the data.

Taken together, these results provide direct evidence that future climate scenarios occupy regions of the feature space that differ from those represented in the Historical climate. The climate generalization problem considered in this work can therefore be interpreted as a genuine out-of-distribution learning problem, where models trained primarily on historical conditions must remain accurate when deployed on climate states characterized by shifted and broadened multivariate distributions.

4 Methods

4.1 Climate Representation Alignment Framework

This work builds upon the Climate-Equivariant Representation Alignment (CERA) framework introduced by Liu and O’Gorman [9]. Rather than proposing a fundamentally new architecture, our objective is to evaluate the validity of several key assumptions underlying CERA in the context of realistic CMIP6 climate simulations. Consequently, we adopt a framework that remains as close as possible to the original CERA design while introducing a limited number of controlled modifications aimed at isolating specific methodological choices.

As discussed in Section 2, the CERA architecture combines three objectives: (i) learning a compact latent representation through an autoencoder, (ii) aligning latent representations originating from different climates, and (iii) optimizing predictive performance on a downstream task. We follow the same general principle. A shared encoder maps multivariate climate patches into a latent representation, while a decoder reconstructs the original input and a predictor estimates the target variable from the latent space. Historical and future-climate samples are processed by the same encoder, thereby enforcing a common representation space across climates.

Following CERA, the latent representation is partitioned into two components. The first component is subjected to an alignment constraint and serves as input to the downstream predictor. The second component remains unconstrained and is only used by the decoder during reconstruction. This design is motivated by the hypothesis that some information may be climate-dependent and useful for reconstruction while not being required for climate-invariant prediction. As in the original framework, reconstruction losses are applied to samples from both climates, whereas supervised prediction losses are computed exclusively on Historical samples.

The overall optimization objective follows the formulation proposed in CERA:

$$L = (1 - \lambda_{\text{rec}} - \lambda_{\text{align}})L_{\text{pred}} + \lambda_{\text{rec}}L_{\text{rec}} + \lambda_{\text{align}}L_{\text{align}},$$

where L_{pred} denotes the supervised prediction loss, L_{rec} the reconstruction loss, and L_{align} the latent alignment loss. The relative weights control the trade-off between predictive performance, reconstruction quality, and climate invariance.

While the original CERA architecture was developed for one-dimensional atmospheric profiles using 1D convolutional layers, the present study operates on two-dimensional climate patches. We therefore replace the original encoder and decoder with convolutional architectures adapted to spatial climate fields. This modification preserves the overall philosophy of CERA while allowing the model to exploit local spatial patterns and neighborhood information contained in the climate patches.

Several alignment strategies are investigated within this common framework. The primary alignment strategy considered in this work is based on the Sliced Wasserstein Distance (SWD). The original CERA framework relies on the Earth Mover’s Distance (EMD) to align latent representations across climates. Since EMD becomes computationally expensive in high-dimensional latent spaces, we replace it with SWD, which provides an efficient approximation while preserving the same underlying objective of distribution matching. Consequently, this variant should be viewed as the closest practical implementation of the original CERA methodology rather than as a conceptual modification of the framework. Throughout the remainder of this work, SWD therefore serves as the reference implementation of climate representation alignment.

To further investigate the role of the alignment mechanism itself, we also consider an adversarial alignment strategy. In this setting, a domain classifier is trained to identify the climate associated with a latent representation, while the encoder is simultaneously optimized to prevent such discrimination. Unlike SWD, this approach does not explicitly match latent distributions

and instead encourages climate invariance through domain confusion. The adversarial framework therefore provides a substantially different realization of climate-invariant learning and allows us to assess whether the benefits observed in CERA originate specifically from distribution matching or more generally from the enforcement of climate-invariant representations.

Finally, we introduce a normalized variant of the Sliced Wasserstein alignment motivated by observations made during the analysis of the learned latent spaces. Preliminary experiments revealed that standard SWD often encourages a global contraction or rescaling of the latent geometry. While this behavior effectively reduces the alignment loss, it may do so by shrinking distances rather than by genuinely improving the relative alignment of climate representations. To mitigate this effect, we normalize latent distances prior to computing the alignment objective, thereby making the loss less sensitive to global geometric rescaling. This variant, referred to as Normalized SWD (SWDN), was introduced both as a potential improvement to the original CERA framework and as a tool for better understanding how alignment constraints shape latent-space geometry.

4.2 Experimental Setups and Baselines

The objective of this study is not only to evaluate the predictive performance of climate representation alignment, but also to understand how the different components of the CERA framework contribute to the emergence of climate-invariant representations. To this end, we consider a series of experimental setups that progressively reconstruct the original CERA architecture, together with several baselines and architectural variants designed to test its underlying assumptions.

Table 1: Summary of the experimental setups considered in this study.

Setup	Autoencoder	Alignment	Predictor	Training Climates
AEh	✓	–	–	Historical
AEhs2	✓	–	–	Hist. + SSP2-4.5
AEhs2s3	✓	–	–	Hist. + SSP2-4.5 + 3-7.0
AEall	✓	–	–	Hist. + SSP2-4.5 + 3-7.0 + 5-8.5
Exp4	✓	SWD	–	Hist. + SSP2-4.5
CERA	✓	SWD	Aligned latent	Hist. + SSP2-4.5
NoAlign	✓	–	Aligned latent	Hist. + SSP2-4.5
Simple	–	–	MLP	Historical
Physical	–	–	MLP + invariants	Historical
ClimaX	Found. model	–	Prediction head	Historical fine-tuning
CERA-Full	✓	SWD	Full latent	Hist. + SSP2-4.5
CERA-Adv	✓	Adversarial	Aligned latent	Hist. + SSP2-4.5
CERA-SWDN	✓	SWDN	Aligned latent	Hist. + SSP2-4.5

Progressive Construction of Climate-Aligned Representations

The first group of experiments aims to isolate the contribution of the different components composing CERA. We begin with a standard autoencoder trained solely through reconstruction losses and progressively introduce additional climates and alignment constraints.

The simplest configuration, denoted *AEh*, is trained exclusively on Historical samples. We then successively increase the diversity of the training data by introducing additional climate scenarios. The resulting models are denoted *AEhs2*, *AEhs2s3*, and *AEall*, corresponding respectively to training on Historical + SSP2-4.5, Historical + SSP2-4.5 + SSP3-7.0, and Historical + SSP2-4.5 + SSP3-7.0 + SSP5-8.5. These experiments allow us to investigate whether exposing a shared encoder to multiple climates is sufficient to naturally reduce climate discrepancies in latent space, even in the absence of any explicit alignment objective.

A second intermediate setup, referred to as *Exp4*, introduces latent alignment while keeping the purely unsupervised autoencoding framework. In this configuration, the model is trained on

Historical and SSP2-4.5 samples and an alignment loss is applied between the latent representations of the two climates. This experiment isolates the effect of explicit climate alignment from that of supervised prediction and allows us to study its direct impact on latent-space organization.

Finally, the complete CERA-like architecture corresponds to the supervised setting described in the previous section. The model combines reconstruction, alignment, and prediction objectives and is trained using Historical labels only, while unlabeled SSP2-4.5 samples are used exclusively for latent alignment.

CERA Baselines

To assess whether climate representation alignment provides a tangible benefit, we compare the CERA-like architecture against several baselines inspired by the original CERA study.

The first baseline, denoted *NoAlign*, reproduces the CERA architecture while removing the alignment objective. The encoder, decoder, latent partitioning, and predictor remain unchanged, allowing the effect of explicit climate alignment to be isolated. This baseline is particularly important because it directly tests whether the performance gains reported by CERA originate from latent alignment itself rather than from the underlying autoencoder architecture.

The second baseline, referred to as *Simple*, consists of a standard multilayer perceptron trained directly on the raw climate patches. No latent representation learning or climate alignment is performed. This setup provides a reference corresponding to a straightforward supervised learning approach and serves as a lower-complexity benchmark.

The third baseline, denoted *Physical*, follows the same prediction framework as the Simple baseline but augments the inputs with manually engineered climate-invariant features inspired by the climate-invariant machine learning framework of Beucler et al. [8]. These features include both physically motivated climate invariants and statistical climate invariants. The physical invariants comprise quantities such as relative humidity, vertical stability indicators, wind shear, and wind magnitude, while the statistical invariants correspond to climatological anomalies computed for each input variable relative to the Historical training climatology. This baseline evaluates whether explicitly prescribing candidate climate-invariant features, derived either from physical knowledge or statistical normalization, can provide benefits comparable to those obtained through automatic invariant discovery.

Finally, we consider the climate foundation model *ClimaX* [2]. Following the protocol proposed by Navarro et al. [7], a prediction head is trained on top of a pretrained ClimaX encoder using Historical data only. This baseline provides a comparison against a state-of-the-art climate foundation model trained on large-scale climate datasets. Comparing ClimaX and CERA allows us to contrast two distinct strategies for climate generalization: large-scale pretraining on diverse climate data versus explicit climate representation alignment.

CERA Variants

Beyond reproducing the original framework, we investigate several variants designed to question specific assumptions underlying CERA.

The first variant, denoted *CERA-Full*, provides the predictor with access to the entire latent representation rather than only the aligned latent dimensions. This experiment questions a central design choice of CERA, namely that prediction should rely exclusively on the climate-aligned component of the latent space. While restricting the predictor to aligned features may reduce the risk of exploiting climate-specific correlations, it may also discard information that remains useful for prediction. CERA-Full therefore investigates the trade-off between enforcing climate invariance and preserving predictive information.

The second variant replaces the SWD alignment loss with an adversarial alignment objective. A climate classifier is trained to identify the climate associated with a latent representation, while the encoder simultaneously attempts to remove this information. Unlike SWD, this approach does not explicitly match latent distributions and therefore provides an alternative mechanism for learning climate-invariant representations.

The final variant, referred to as *CERA-SWDN*, introduces a normalized version of the Sliced Wasserstein Distance. This modification was motivated by preliminary observations indicating that standard SWD can encourage global geometric rescaling of the latent space in addition to distribution alignment. By normalizing latent distances prior to alignment, the objective becomes less sensitive to global contractions and provides a more direct measure of relative geometric similarity between climates.

Together, these experiments form a hierarchy of increasingly sophisticated models, ranging from standard autoencoders to climate-aligned architectures and foundation models. This design allows us to separately evaluate the effects of climate diversity, latent alignment, prediction constraints, and alignment mechanisms on both predictive performance and latent-space organization.

4.3 Latent Space Analysis

A central objective of this work is to understand how climate representation alignment influences the organization of the learned latent space. While predictive performance provides an indirect measure of generalization, it does not reveal how different climates are represented internally by the models. To address this question, we perform a systematic analysis of the latent representations learned by the various architectures introduced in the previous section.

Latent Representations Considered

Latent-space analyses are conducted on all models that learn an explicit intermediate representation. This includes the autoencoder-based experiments (*AEh*, *AEhs2*, *AEhs2s3*, *AEall*, and *Exp4*), the complete CERA-like architecture, its variants (*CERA-Full*, *CERA-Adv*, and *CERA-SWDN*), the *NoAlign* baseline, and the pretrained *ClimaX* model. Depending on the architecture, analyses are performed on either the full latent representation or on the aligned component when specifically studying the effect of climate representation alignment.

Visualization of Latent Spaces

We begin by qualitatively investigating latent-space organization through low-dimensional visualizations. Principal Component Analysis (PCA) is used to characterize the dominant directions of variability and provide a global view of the latent geometry, while t-distributed Stochastic Neighbor Embedding (t-SNE) is employed to reveal local neighborhood structures that may not be visible through linear projections. Climate samples from different scenarios are projected into these low-dimensional spaces and visualized jointly. These representations provide an intuitive first assessment of the degree of overlap, separation, or alignment between climates before introducing quantitative measures.

Cross-Climate Distances

To quantify the differences observed in the visualizations, we compare latent distributions across climates using several geometric metrics. In particular, we compute distances between latent centroids as well as Wasserstein-based measures that characterize discrepancies between the full distributions. These metrics provide a quantitative assessment of climate alignment and allow latent spaces learned by different architectures to be compared on a common basis. Because latent representations may differ in scale across models, distance-based analyses are systematically normalized to facilitate meaningful comparisons of latent geometry.

Dimensionality and Geometry

Beyond distributional alignment, we investigate the internal structure of the learned representations. In particular, we analyze the effective rank of the latent space as a measure of the dimensionality effectively utilized by the model. Effective rank provides insight into whether latent representations occupy a large portion of the available latent space or instead collapse onto a smaller number of dominant directions. Combined with the geometric metrics described above, these analyses are complemented by measurements of the latent-space root mean square (RMS), which provide a quantitative estimate of global geometric rescaling. Together, these metrics help characterize how alignment constraints influence the structure, scale, and complexity of the learned representations.

Climate Separability

Finally, we evaluate the extent to which climate information remains encoded in the latent space. To this end, classifiers are trained to predict the climate scenario associated with a latent representation. If climate labels can be recovered accurately, the latent space still contains climate-specific information; conversely, poor classification performance suggests that the learned representations have become more climate-invariant. Together with the geometric analyses described above, this provides a direct assessment of whether alignment methods effectively remove climate-dependent information from the latent space. However, demonstrating climate invariance is only a first step. Even if climate information is no longer accessible, it remains unclear whether the resulting representations capture meaningful physical invariants or merely reflect statistical artifacts induced by the learning process. Addressing this question requires a dedicated analysis of the physical content of the latent space.

4.4 Physical Invariant Analysis

The latent-space analyses described in the previous section allow us to assess whether climate information remains encoded in the learned representations. However, reduced climate separability does not necessarily imply that the model has learned meaningful climate invariants. A latent space may become climate-invariant while still organizing samples according to spurious statistical structures unrelated to physically relevant processes. To evaluate the central hypothesis underlying climate-invariant learning, we therefore investigate whether the learned representations encode physically meaningful invariant quantities and to what extent the observed latent organization is consistent with these invariants.

Candidate Physical Invariants

We begin by considering a set of physically motivated climate invariants derived from atmospheric thermodynamics and dynamics. These include relative humidity at multiple pressure levels, vertical stability indicators, wind shear, and wind magnitude. In addition, we consider statistical climate invariants in the form of climatological anomalies computed relative to the Historical training climatology. Although these statistical quantities are not analyzed in detail, they provide a useful reference for comparison with physically motivated invariants.

Invariant Visualization and Probing

As a first step, we investigate whether candidate invariants are organized within the latent space. Low-dimensional latent representations obtained through t-SNE are colored according to the value of each invariant in order to identify potential large-scale structures associated with physical quantities. These visualizations are complemented by quantitative probing analyses. Linear probes are first used to evaluate the extent to which each invariant can be recovered from the latent representation. We then employ a k-nearest-neighbor probing strategy, in which the invariant associated with a sample is predicted from the invariants of its nearest latent neighbors. High probing performance indicates that samples with similar invariant values occupy nearby

regions of latent space, suggesting that the representation has learned an organization consistent with the underlying physical quantity.

Neighborhood Consistency Across Climates

To evaluate whether latent alignment preserves physically meaningful relationships across climates, we analyze pairs of neighboring samples in latent space, where one sample originates from the Historical climate and the other from a future SSP scenario. For each pair, differences in the candidate invariants are computed and compared to their latent proximity. This analysis provides a direct measure of cross-climate consistency: if latent alignment successfully captures climate-invariant structures, samples that become neighbors in latent space should exhibit similar values of the underlying physical invariants. Conversely, large invariant discrepancies between neighboring samples would suggest that climate alignment does not correspond to meaningful physical similarity.

Invariant Discovery

Finally, we investigate whether the learned representations encode invariant relationships that are not explicitly included among the candidate invariants. To this end, we formulate invariant discovery as a search for quantities that remain approximately constant between neighboring samples originating from different climates.

Two complementary approaches are considered. First, sparse linear combinations of the original climate variables are identified using LASSO regression. The optimization is performed on differences between neighboring cross-climate samples, with the objective of finding combinations of variables whose variations are minimized. Second, symbolic regression is employed to search for simple nonlinear expressions that remain approximately invariant across aligned samples.

Together, these analyses provide a systematic framework for evaluating whether climate-aligned representations merely reduce climate separability or genuinely capture physically meaningful invariant structures. They also enable the discovery of candidate invariants directly from data, thereby offering a complementary perspective to manually engineered climate-invariant features.

5 Climate Generalization Experiments

5.1 Temperature Prediction

We first evaluate the ability of the different architectures to predict near-surface air temperature (**tas**) under increasingly severe climate distribution shifts. Temperature prediction constitutes a relatively simple benchmark compared to precipitation, as surface temperature is strongly constrained by large-scale thermodynamic relationships and remains highly correlated with several of the input variables available to the models. Consequently, this task provides an initial assessment of climate generalization before moving to more challenging prediction targets.

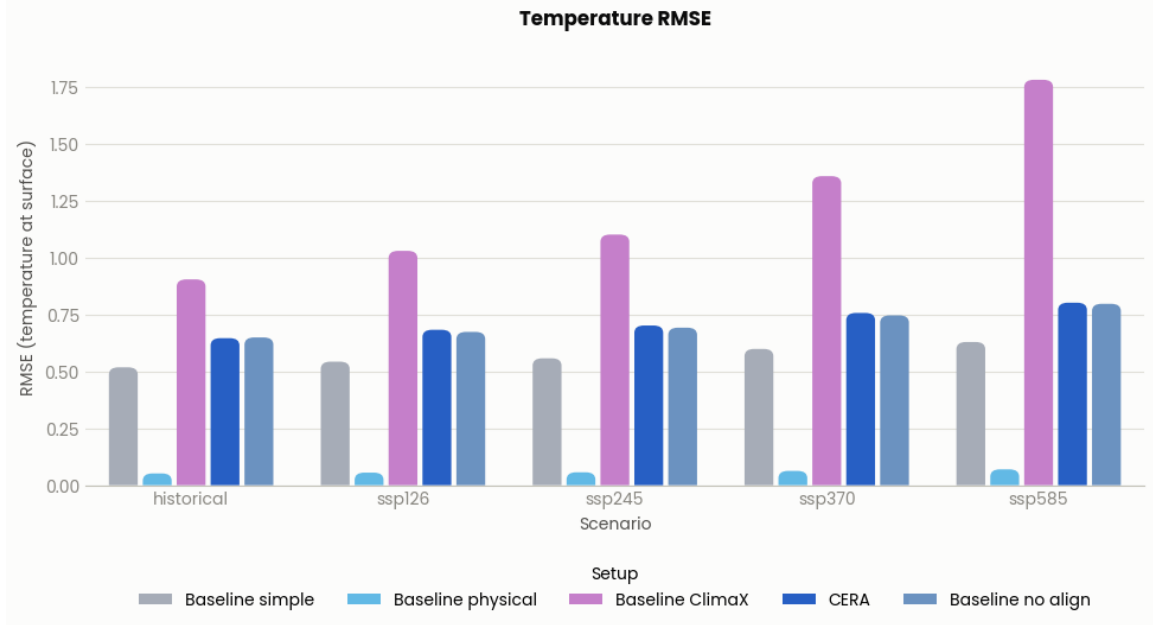


Figure 7: Temperature prediction RMSE for each climate scenario and model. Lower values indicate better predictive performance.

Figure 7 presents the RMSE obtained by the different models on the temperature prediction task. Overall, all architectures achieve relatively low prediction errors across all climate scenarios, confirming that surface temperature is substantially easier to predict than variables such as precipitation. Even under the strongest warming scenario (SSP585), the degradation remains moderate for most models.

Among all methods, the Physical baseline clearly achieves the best absolute performance. This result is not entirely surprising, as several of the handcrafted physical invariants introduced in this baseline are directly related to the thermal structure of the atmosphere, including relative humidity, lapse rates, and temperature gradients. As a consequence, this baseline benefits from additional information that is particularly relevant for temperature prediction. The exceptionally low RMSE obtained by this model should therefore be interpreted with caution, as the task is inherently favorable to the selected handcrafted features.

The Simple baseline also performs remarkably well despite its architectural simplicity, suggesting that a large fraction of the information required to predict surface temperature is already present in the raw atmospheric variables. Interestingly, CERA and the NoAlign baseline exhibit nearly identical performance across all climate scenarios. This observation suggests that explicit latent-space alignment provides limited benefits for this particular task. Since the statistical relationships governing temperature remain relatively stable across climates, enforcing climate-invariant representations may simply be less critical than for more nonlinear prediction problems.

Finally, ClimaX performs noticeably worse than the other approaches. While foundation models are generally expected to provide strong transfer capabilities, the pretrained representations learned by ClimaX do not appear particularly advantageous for this specific temperature prediction task.

To better isolate the effect of climate distribution shift, Figure 8 reports the RMSE increase relative to the Historical climate.

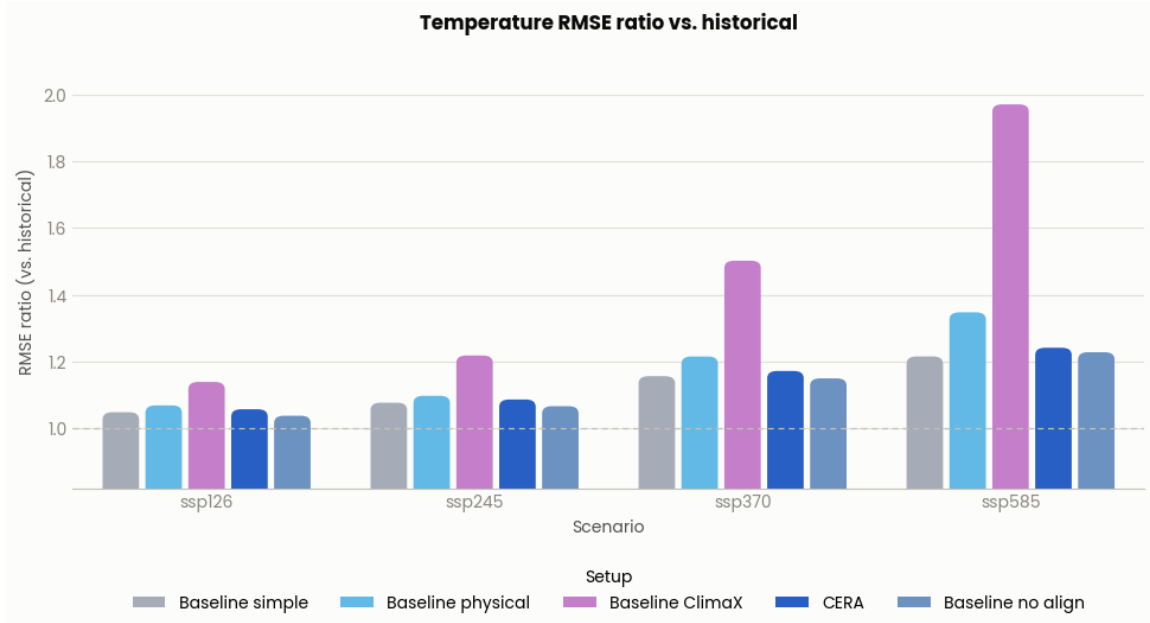


Figure 8: Relative increase in temperature prediction RMSE compared to the Historical climate.

All models exhibit a gradual increase in prediction error as the climate becomes warmer, reflecting the growing distribution shift between the training and evaluation climates. However, the magnitude of this degradation remains relatively limited compared to what will be observed later for precipitation prediction.

The Physical baseline remains remarkably stable across all scenarios, further illustrating the strong predictive power of the handcrafted thermodynamic quantities for this task. In contrast, ClimaX experiences the largest relative degradation as warming increases, despite being pre-trained on a broad collection of climate datasets. Interestingly, both CERA and NoAlign display nearly identical robustness profiles, with only marginal differences between the two methods. This result again suggests that latent alignment is not a major factor for temperature prediction and that most of the required information can already be extracted without explicitly enforcing climate invariance.

These observations motivate the study of a more challenging and physically nonlinear target: precipitation prediction.

5.2 Precipitation Prediction

We now consider precipitation prediction, which constitutes a substantially more challenging climate generalization problem than temperature prediction. Unlike surface temperature, precipitation results from highly nonlinear interactions between atmospheric thermodynamics, moisture transport, convection, and large-scale circulation. Moreover, precipitation distributions are strongly skewed, intermittent, and known to undergo significant modifications under climate change. As a consequence, this task provides a more demanding benchmark for evaluating the ability of learned representations to generalize across climates.

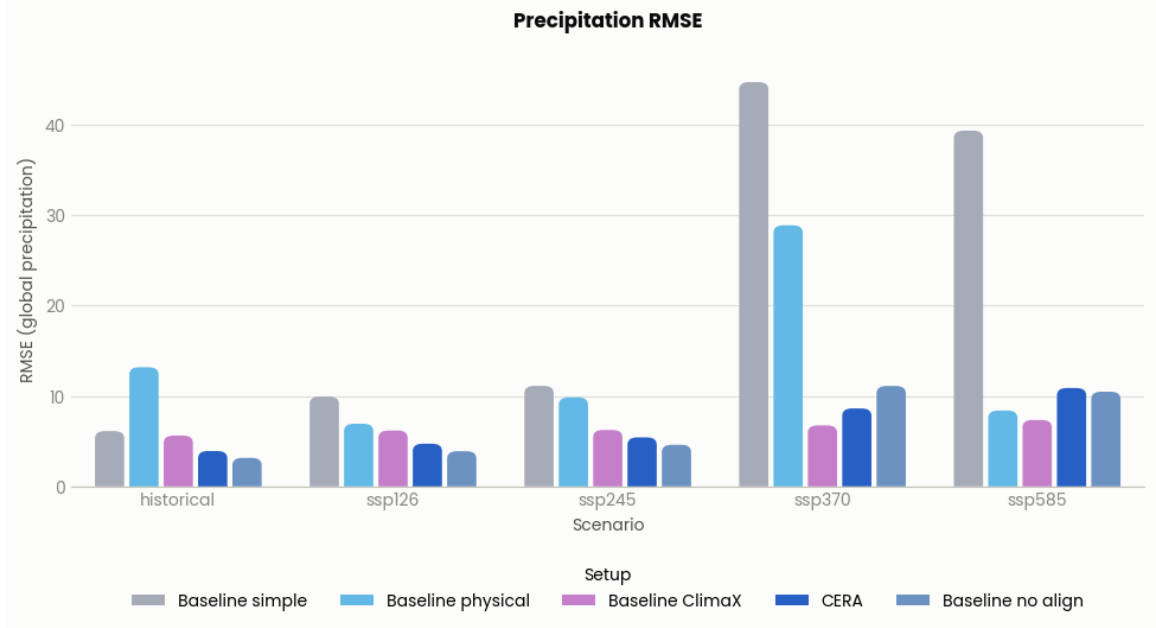


Figure 9: Precipitation prediction RMSE for each climate scenario and model.

Figure 9 reports the precipitation prediction RMSE obtained by the different architectures. In contrast to the temperature prediction task, substantial differences emerge between models as the climate becomes warmer. While all approaches achieve relatively similar performance on the Historical, SSP126 and SSP245 climates, their behavior diverges considerably under stronger distribution shifts.

The Simple baseline exhibits the largest degradation. Its RMSE increases dramatically in SSP370 and SSP585, indicating that statistical relationships learned from the Historical climate do not transfer reliably to warmer climates. This behavior is characteristic of a severe out-of-distribution generalization failure and illustrates the difficulty of precipitation prediction under climate change.

Introducing physically motivated invariants significantly improves robustness. The Physical baseline consistently outperforms the Simple baseline and exhibits a much smaller degradation across scenarios. This result supports the idea that physically meaningful quantities provide information that remains relevant across climates.

Among the learned approaches, both CERA and NoAlign achieve substantially lower errors than the Simple baseline and remain competitive even under the strongest warming scenarios. Interestingly, the difference between CERA and NoAlign remains relatively small across all climates. Although CERA generally achieves slightly lower errors, the magnitude of the improvement is considerably smaller than the gap separating both methods from the Simple baseline. This observation suggests that factors other than explicit latent alignment may contribute significantly to the observed robustness.

Finally, Climax achieves the best overall performance. Its RMSE remains remarkably low even in SSP370 and SSP585, highlighting the strong transfer capabilities provided by large-scale pretraining. The ability of Climax to maintain good performance under substantial climate shifts suggests that its pretrained representations capture structures that remain relevant across a wide range of climate conditions.

To better isolate the impact of climate change on model performance, Figure 10 presents the RMSE increase relative to the Historical climate.

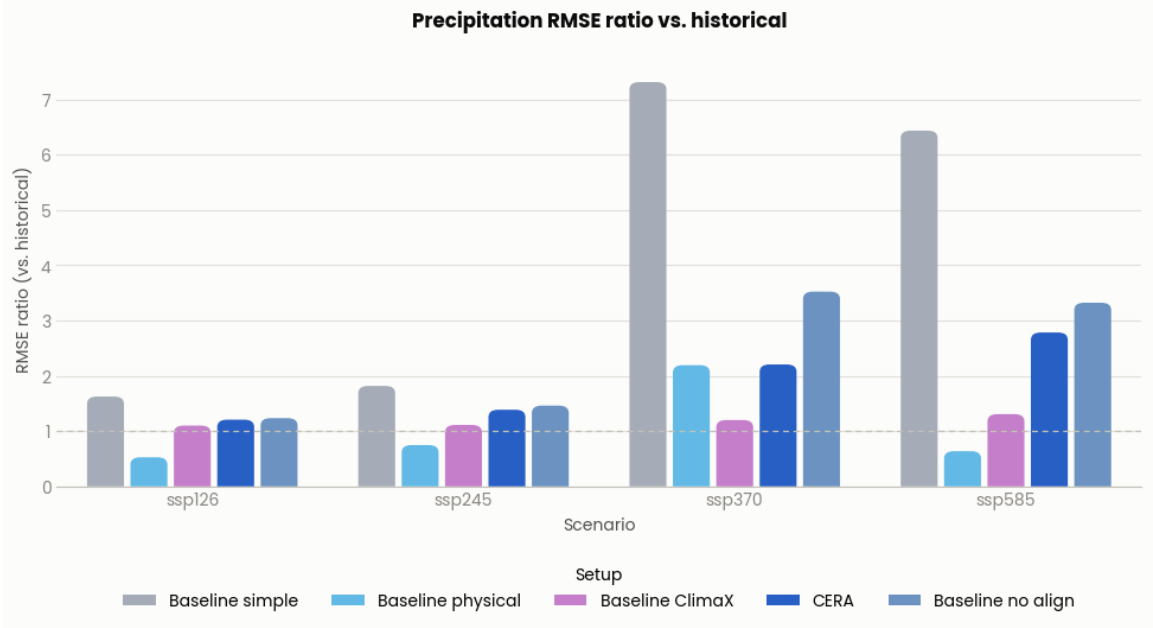


Figure 10: Relative increase in precipitation prediction RMSE compared to the Historical climate.

The contrast between the different approaches becomes even clearer when considering relative degradation. The Simple baseline experiences a dramatic increase in error, reaching more than six times its Historical RMSE in SSP370 and SSP585. Such a deterioration confirms that precipitation prediction is highly sensitive to distribution shift.

The Physical baseline exhibits substantially better robustness, while Climax remains the most stable model overall. Remarkably, the relative degradation of Climax remains close to unity even under the strongest warming scenarios, indicating that its performance remains nearly unchanged despite large changes in climate statistics.

CERA occupies an intermediate position. Although its error increases significantly in warmer climates, the degradation remains far smaller than that observed for the Simple baseline. However, the similarity between CERA and NoAlign is again striking. Both architectures follow almost identical degradation trends across climates, raising important questions regarding the precise role played by explicit latent alignment in the climate generalization process.

While global RMSE provides a useful summary of predictive performance, it does not fully characterize precipitation prediction. Because most precipitation events are weak or absent, a model may achieve a satisfactory RMSE while failing to reproduce the most intense events. We therefore investigate the behavior of the models on precipitation extremes.

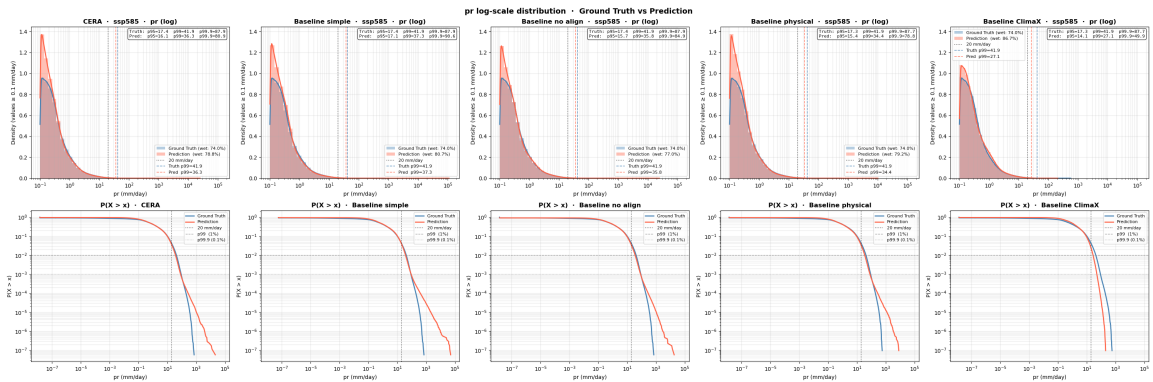


Figure 11: Precipitation distributions and exceedance probabilities for SSP585. The upper row compares predicted and observed distributions, while the lower row shows exceedance probabilities $P(X > x)$ on a logarithmic scale.

Figure 11 compares the predicted and observed precipitation distributions for SSP585, the most challenging climate considered in this study. Several important differences emerge.

The Simple baseline struggles to reproduce the tail of the precipitation distribution and substantially overestimates the frequency of extreme events. This behavior becomes particularly visible in the exceedance probability curves, where the predicted tail remains significantly heavier than the observed one.

Both CERA and NoAlign provide a much better approximation of the true precipitation distribution. Their exceedance curves remain considerably closer to the ground truth than those obtained with the Simple baseline, indicating a better representation of extreme precipitation behavior. Nevertheless, the two models remain almost indistinguishable from one another, suggesting that the improvements arise primarily from the learned latent representation itself rather than from the explicit alignment constraint.

ClimaX achieves the best agreement with the observed distribution. Both the probability density function and the exceedance curves closely match the ground truth, demonstrating a superior ability to capture the statistics of rare and intense precipitation events.

To quantify these observations, we compute the ratio between the RMSE obtained on extreme precipitation events ($pr > 10$ mm/day) and the global RMSE.

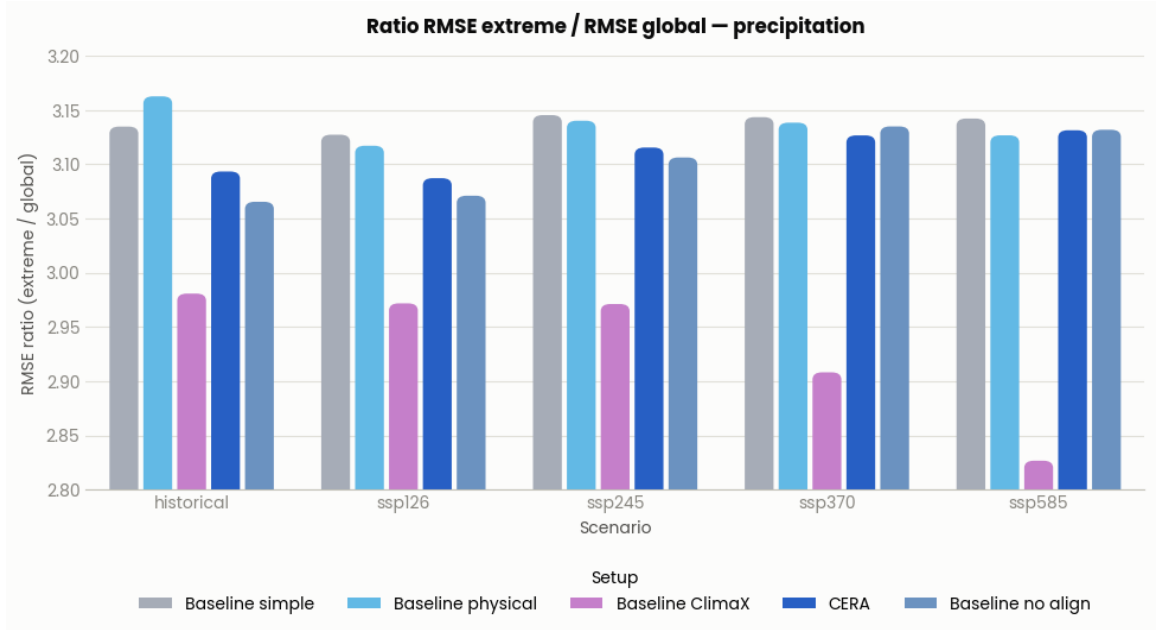


Figure 12: Ratio between extreme-event RMSE ($pr > 10$ mm/day) and global precipitation RMSE. Lower values indicate better relative performance on extreme events.

The results confirm the conclusions drawn from the distributional analysis. ClimaX consistently achieves the lowest ratio across all climates, indicating superior performance on extreme precipitation events. CERA and NoAlign form a second group with nearly identical behavior, both outperforming the Simple baseline. The Physical baseline improves over the Simple baseline but remains less effective than the latent-representation approaches.

Summary

Precipitation prediction reveals strong climate distribution-shift effects and therefore constitutes a far more discriminative benchmark than temperature prediction. CERA substantially improves over simple supervised learning approaches and remains competitive under severe warming scenarios. However, the consistently small gap between CERA and NoAlign suggests that

the precise contribution of latent alignment remains unclear. At the same time, ClimaX achieves the best overall robustness and the best representation of precipitation extremes, indicating that large-scale pretraining may also promote the learning of transferable climate representations. These observations motivate a deeper investigation of the latent spaces learned by the different architectures.

5.3 Masked Reconstruction

As a final benchmark, we consider a masked reconstruction task in which a subset of the input variables is hidden and must be reconstructed from the remaining spatial and multivariate context. Unlike temperature and precipitation prediction, this task does not focus on a specific climate variable and therefore provides a complementary assessment of representation quality. Details of the masking procedure are provided in the Appendix.

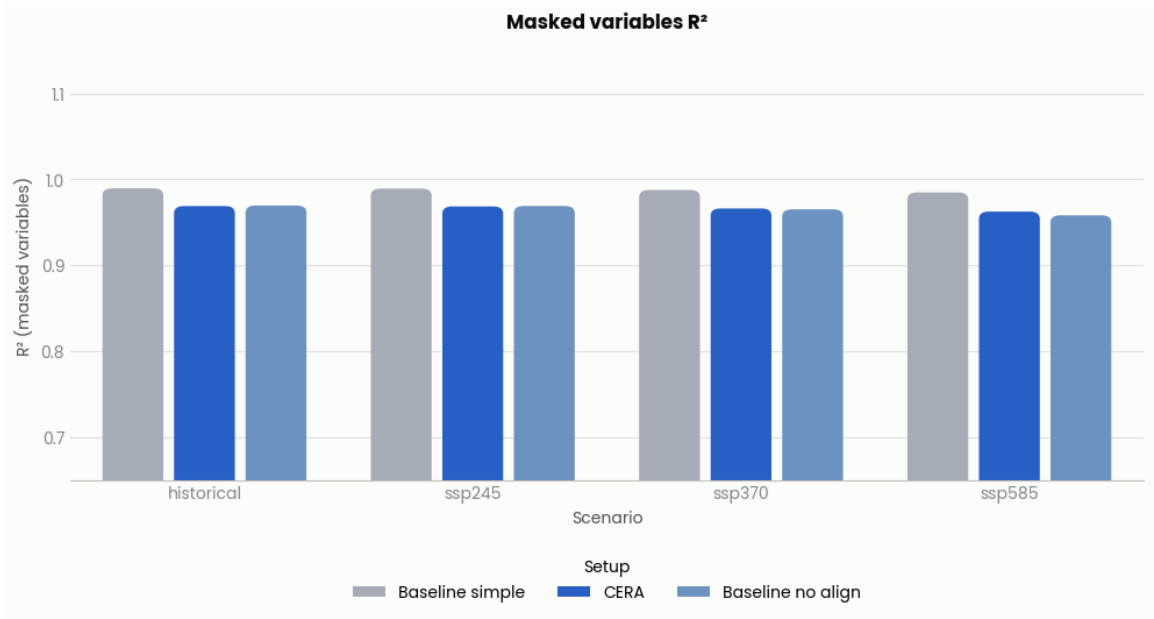


Figure 13: Average reconstruction R^2 for masked variables across climate scenarios. Results are averaged over all reconstructed variables.

Figure 13 reports the average reconstruction R^2 obtained by the Simple baseline, NoAlign, and CERA. All models achieve very high scores, with R^2 values consistently above 0.95 across all climate scenarios. This indicates that the atmospheric state can be reconstructed accurately from the available contextual information and suggests that the task is substantially easier than precipitation prediction.

The differences between architectures remain small. The Simple baseline consistently achieves the highest reconstruction accuracy, while CERA and NoAlign exhibit nearly identical performance. This behavior suggests that the additional constraints introduced by the autoencoder and alignment objectives do not provide a clear advantage for reconstruction. On the contrary, forcing information through a compressed latent representation may slightly reduce reconstruction accuracy compared to a model optimized solely for this objective.

Masked reconstruction provides a useful complementary benchmark but appears insufficiently challenging to clearly discriminate between climate-alignment strategies. In contrast, precipitation prediction exhibits strong distribution-shift effects and much larger differences between architectures. For this reason, the remainder of this section focuses exclusively on precipitation prediction in order to investigate whether modifications of the CERA framework can further improve climate generalization.

5.4 Analysis of CERA Variants on Precipitation Prediction

The previous experiments demonstrated that CERA achieves competitive performance on precipitation prediction under climate distribution shifts. However, they also revealed two important observations. First, the performance gap between CERA and NoAlign remains relatively small, raising questions about the precise contribution of the alignment mechanism. Second, several assumptions underlying the original CERA framework remain largely untested. We therefore investigate three variants of CERA designed to challenge specific architectural choices and alignment strategies introduced in the original framework.

The first variant, *CERA-Full*, provides the predictor with access to the entire latent representation rather than only the aligned latent dimensions. The second variant, *CERA-Adv*, replaces the distribution-matching objective with an adversarial alignment strategy. Finally, *CERA-SWDN* introduces a normalized version of the Sliced Wasserstein alignment loss designed to mitigate the geometric rescaling effects observed with standard SWD alignment.

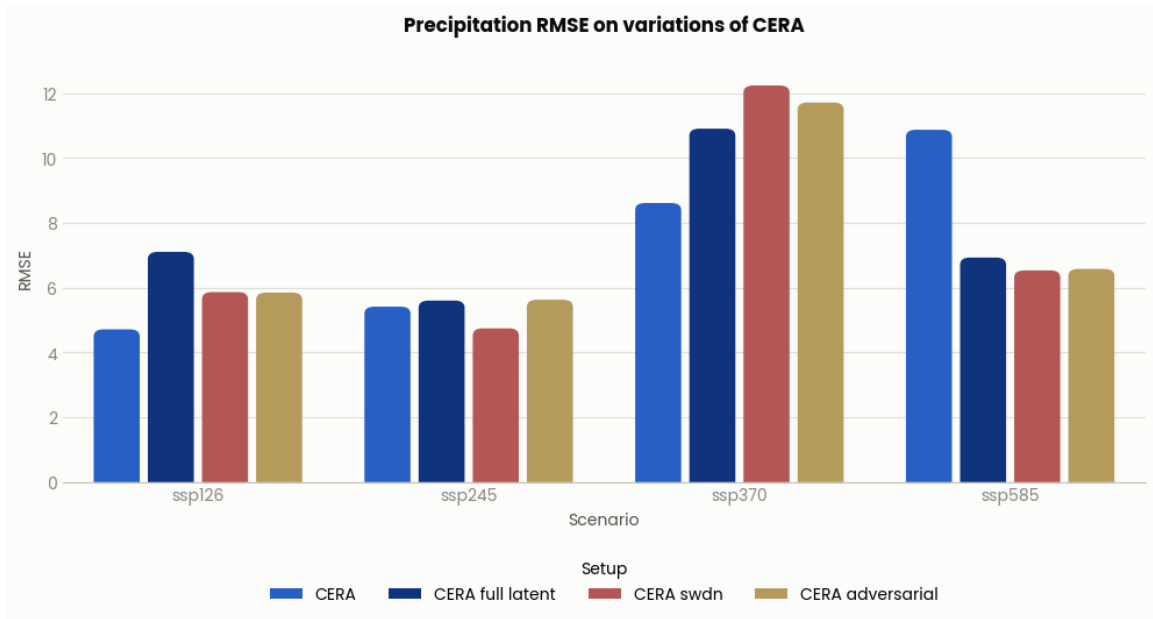


Figure 14: Precipitation prediction RMSE for the different CERA variants across climate scenarios.

Figure 14 compares the precipitation prediction RMSE obtained by the different variants. For the Historical and SSP245 climates, all architectures exhibit nearly identical performance, suggesting that the proposed modifications have little impact when the distribution shift remains limited. More substantial differences emerge under stronger climate changes.

Interestingly, the original CERA architecture does not consistently outperform its variants. While it achieves slightly lower errors in some intermediate climates, all three variants become competitive or superior under SSP585, the strongest warming scenario considered in this study. This result indicates that the original design choices of CERA are not uniquely optimal and that alternative approaches can achieve comparable levels of climate robustness.

Among the variants, *CERA-Full* performs particularly well under large distribution shifts. Allowing the predictor to access the entire latent representation does not appear to harm generalization and may even become beneficial when the target climate departs substantially from the training climate. This observation suggests that completely restricting the predictor to climate-aligned dimensions may remove information that remains useful for prediction.

Both *CERA-Adv* and *CERA-SWDN* also achieve strong performance under SSP585. The success of the adversarial approach indicates that explicit distribution matching is not the only

mechanism capable of promoting transferable climate representations. Similarly, the strong performance of SWDN suggests that the geometric rescaling effects induced by standard SWD alignment may not be necessary to achieve robust climate generalization.

To better isolate the impact of climate change, Figure 15 reports the RMSE increase relative to the Historical climate.

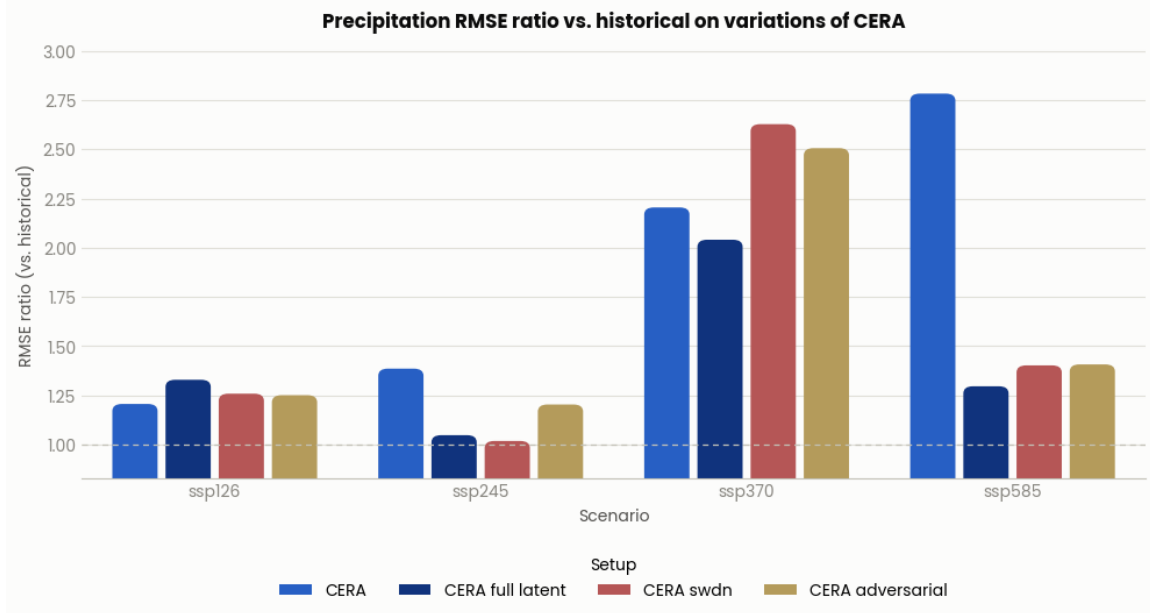


Figure 15: Relative increase in precipitation prediction RMSE compared to the Historical climate for the different CERA variants.

The relative degradation curves confirm the conclusions drawn from the absolute RMSE values. All variants exhibit broadly similar behavior under moderate climate shifts, but important differences emerge for SSP585. In particular, *CERA-Full*, *CERA-Adv*, and *CERA-SWDN* all display a smaller degradation than the original CERA architecture. Although the magnitude of the improvement remains moderate, the consistency of this trend suggests that the modifications introduced by the variants improve robustness under the largest distribution shifts.

These results are particularly interesting in light of the hypotheses discussed in the previous sections. The strong performance of *CERA-Full* challenges the assumption that prediction should rely exclusively on climate-aligned latent dimensions. Likewise, the success of *CERA-Adv* indicates that learning climate-invariant representations does not necessarily require explicit distribution matching. Finally, the behavior of *CERA-SWDN* suggests that the standard SWD objective may encourage geometric transformations that are not directly related to climate invariance and whose impact deserves further investigation.

Summary

Overall, the proposed variants achieve performance comparable to, and in the strongest climate shifts sometimes better than, the original CERA framework. These results suggest that the mechanisms responsible for climate generalization are more subtle than latent distribution alignment alone. In particular, the improvements obtained with *CERA-Full*, *CERA-Adv*, and *CERA-SWDN* motivate a deeper investigation of the latent representations learned by these models. Understanding how the different alignment strategies organize the latent space and what information they preserve or discard is therefore the focus of the next section.

6 Latent Representation Analysis

6.1 Latent Reconstruction Quality

Before analyzing the geometry and climate invariance of the latent representations, it is first necessary to verify that these representations actually encode meaningful information about the input state. Indeed, conclusions drawn from latent-space analyses would be difficult to interpret if the latent representations suffered from severe information loss or latent collapse.

To assess the amount of information preserved by each representation, we evaluate the reconstruction performance of the autoencoder component. For each setup, the decoder is tasked with reconstructing the original climate variables from the latent representation, and reconstruction quality is measured using the coefficient of determination (R^2). Detailed reconstruction scores for all variables are reported in the Appendix. Table 2 summarizes the average reconstruction performance across variables.

Table 2: Average reconstruction R^2 obtained by the different latent representations.

Setup	AEh	AEhs2	AEhs2s3	AEall	Exp4
Mean R^2	-0.14	0.95	0.95	0.96	0.98

Setup	CERA	NoAlign	CERA-Full	SWDN	Adv.
Mean R^2	0.96	0.96	0.98	0.98	0.98

Most setups achieve reconstruction scores above $R^2 = 0.95$, indicating that the latent representations preserve a large fraction of the information contained in the original climate fields. This result provides an important validation of the subsequent latent-space analyses: the latent representations are not merely low-dimensional projections, but retain sufficient information to accurately reconstruct the atmospheric state.

One notable exception is the *Exp3-AEh* setup, which exhibits negative reconstruction scores for several variables and a strongly negative average R^2 . This behavior is consistent with a severe latent collapse, suggesting that the autoencoder trained exclusively on the Historical climate failed to learn a useful representation. In contrast, introducing additional climates during training immediately restores high reconstruction performance, with all subsequent Exp3 configurations achieving reconstruction scores above 0.95.

Overall, these results indicate that, with the exception of *AEh*, the latent representations studied in the remainder of this section contain substantial and meaningful information about the climate state. Consequently, differences observed in latent geometry, dimensionality, or climate separability can be interpreted as genuine properties of the learned representations rather than artifacts of information loss. In contrast, the severe reconstruction failure observed for *AEh* suggests a latent collapse, implying that the corresponding latent space contains little useful information and should therefore be interpreted with caution in the analyses that follow.

6.2 Geometry of Climate-Aligned Representations

Having established that most latent representations preserve a substantial amount of information about the climate state, we now investigate how this information is organized geometrically. The primary objective of climate representation alignment is to reduce discrepancies between climates in latent space while preserving predictive information. We therefore analyze three complementary aspects of latent geometry: cross-climate distances, latent-space dispersion, and effective dimensionality.

6.2.1 Cross-climate distances

We first quantify the degree of alignment between climates using two complementary metrics. The centroid distance measures the separation between the mean latent representations of different climates, while the Sliced Wasserstein Distance (SWD) compares the full latent distributions. In both cases, distances are computed between the Historical climate and each SSP scenario.

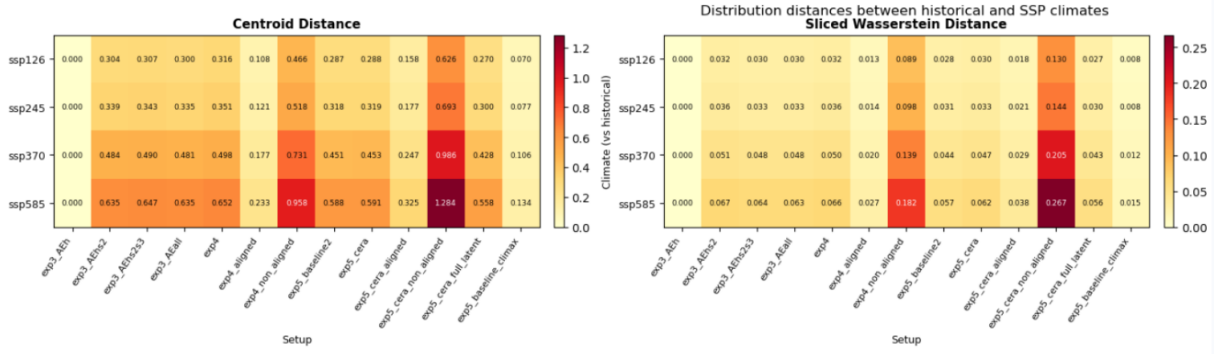


Figure 16: Cross-climate centroid distances (left) and Sliced Wasserstein distances (right) computed between Historical and SSP climates for the different latent representations. Lower values indicate stronger alignment between climates.

Several trends emerge from Figure 16. First, the latent collapse observed for AEh is immediately visible: all distances are essentially zero because the representation contains almost no meaningful information. Consequently, this setup is not considered further in the interpretation of latent geometry.

Among the remaining setups, a progressive reduction of cross-climate distances can already be observed within the Exp3 experiments. As additional climates are introduced during autoencoder training, the Wasserstein distance decreases, indicating that simply exposing the encoder to multiple climates naturally encourages the emergence of more similar representations.

A much stronger effect appears when explicit alignment is introduced. The aligned component of Exp4 exhibits substantially smaller distances than its non-aligned counterpart, confirming that the alignment loss effectively brings climate distributions closer together in latent space. The same behavior is observed for CERA: the aligned latent dimensions display markedly lower centroid and Wasserstein distances than the NoAlign baseline.

Interestingly, ClimaX achieves the smallest cross-climate distances among all meaningful representations. Despite not relying on any explicit alignment mechanism, its pretrained representations appear remarkably stable across climates. This observation is consistent with the strong out-of-distribution performance observed in the previous section and suggests that large-scale pretraining may naturally encourage the emergence of climate-invariant structures.

Overall, these results confirm that the alignment objectives successfully reduce geometric discrepancies between climates and that this effect is already partially induced by multi-climate representation learning.

6.2.2 Latent-space dispersion

Reducing distances between climates does not necessarily imply that meaningful climate-invariant representations have been learned. In principle, an alignment loss could decrease distributional discrepancies simply by shrinking the latent space, thereby reducing all distances simultaneously. To investigate this possibility, we measure the root-mean-square (RMS) dispersion of each latent representation.

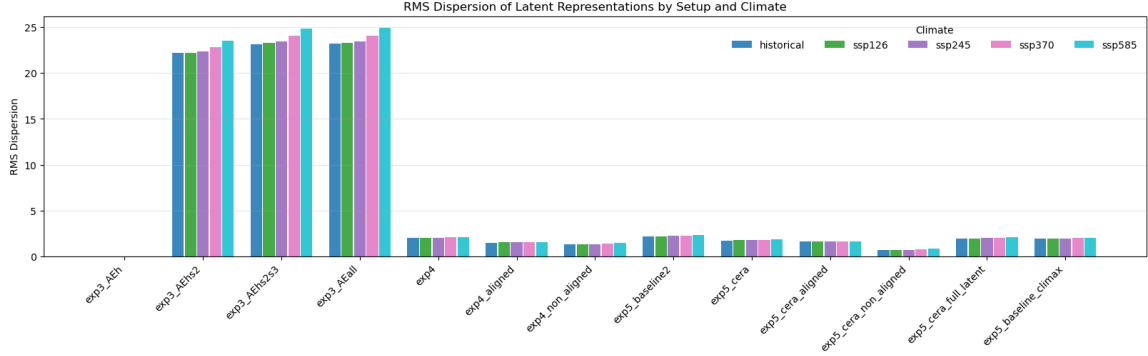


Figure 17: RMS dispersion of latent representations across climates. Lower values correspond to more compact latent spaces.

Figure 17 reveals a striking effect. The Exp3 representations occupy a relatively large volume in latent space, with larger values of RMS. In contrast, introducing explicit alignment causes a dramatic contraction of the latent geometry. The aligned representations of Exp4, CERA, and their variants exhibit much smaller values of RMS.

This result indicates that latent alignment is not achieved solely through a rearrangement of the representation geometry. Instead, the optimization process also performs a substantial global rescaling of the latent space. Such behavior is expected to some extent since reducing distances between distributions naturally encourages more compact representations. However, excessive compression may become problematic if it removes useful information or artificially satisfies the alignment objective without learning genuinely invariant structures.

The observation of this strong geometric rescaling directly motivated the introduction of the SWDN variant presented in Section IV. By normalizing latent distances before alignment, SWDN aims to reduce the incentive for such global contractions and force the model to align climates through their relative geometry rather than their absolute scale.

Importantly, the reconstruction results presented previously demonstrate that this contraction does not lead to a complete latent collapse. Most aligned representations still preserve sufficient information to reconstruct the original climate fields accurately. Nevertheless, the question remains whether the information is distributed differently across latent dimensions after alignment.

6.2.3 Effective dimensionality

To further investigate how information is organized within the latent space, we compute the effective rank of each representation. Effective rank provides a measure of the number of dimensions that contribute meaningfully to the representation and can therefore reveal whether alignment compresses information into a smaller subset of latent directions.

Figure 18 shows that substantial differences exist in how the various models utilize their latent dimensions. Since the latent spaces considered in this study do not all share the same dimensionality, effective-rank values should be interpreted relative to the size of the corresponding latent space rather than in absolute terms.

The aligned and non-aligned components of Exp4 provide an instructive example. Although the aligned representation exhibits a larger absolute effective rank than the non-aligned component, it also operates in a substantially larger latent space. Both representations therefore compress information into a restricted set of dominant directions, but the aligned component distributes information across a richer set of latent features while maintaining strong climate alignment.

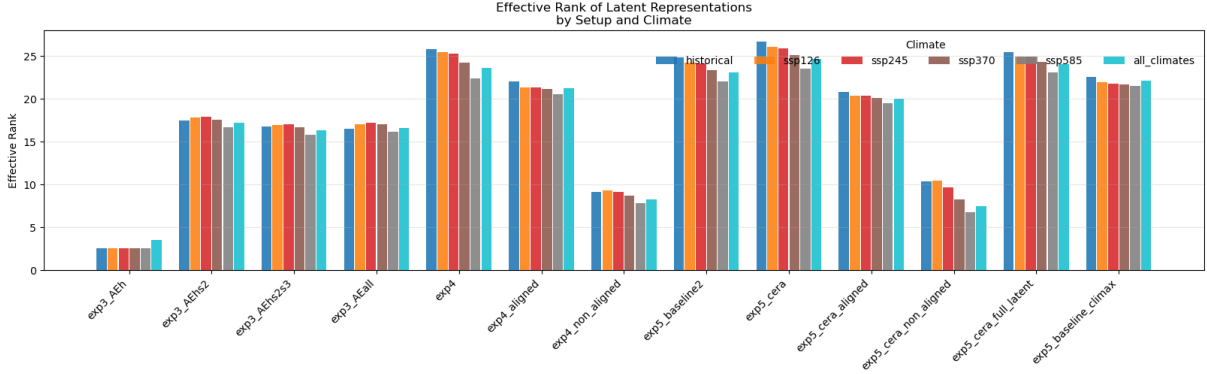


Figure 18: Effective rank of latent representations across climates and experimental setups.

A similar behavior is observed for CERA and its variants. The aligned latent representations typically exhibit effective ranks around 20–21 dimensions for a 48-dimensional latent space, indicating that only part of the available representational capacity is actively used. This suggests that climate alignment promotes a more structured organization of information rather than a uniform use of all latent dimensions.

Interestingly, ClimaX achieves an effective rank comparable to that of CERA despite operating in a larger 64-dimensional latent space. Relative to its latent dimensionality, ClimaX therefore concentrates information into an even smaller fraction of the available representation. This observation suggests that the emergence of compact low-dimensional structures may be a general characteristic of representations that generalize well across climates, regardless of whether climate invariance is obtained through explicit alignment or large-scale pretraining.

Taken together, these results indicate that climate alignment does not simply reduce cross-climate distances. It also reorganizes how information is distributed within the latent space, concentrating predictive information into a limited number of dominant directions while preserving reconstruction quality.

Summary

Taken together, these analyses reveal that climate alignment successfully reduces geometric discrepancies between climates. However, this alignment is accompanied by substantial modifications of latent geometry, including a strong contraction of the latent space and a moderate reduction in effective dimensionality. The resulting representations remain informative despite this compression, suggesting that alignment does not simply collapse the latent space. Having established that climates become geometrically closer while preserving useful information, the next question is whether these aligned representations actually become climate-invariant. We address this issue directly in the following section by analyzing climate separability within the latent space.

6.3 Climate Invariance of the Latent Space

We now evaluate whether the learned representations truly become climate-invariant, by directly analyzing their climate separability.

Figure 19 presents t-SNE visualizations of the latent spaces colored by climate scenario. CERA exhibit substantially stronger mixing between climates than its non-aligned counterparts. The NoAlign baseline displays more localized climate-dependent structures, suggesting that climate information remains more accessible. Interestingly, ClimaX also exhibits strong climate mixing despite the absence of any explicit alignment objective, indicating that large-scale pre-training may naturally encourage climate-invariant representations.

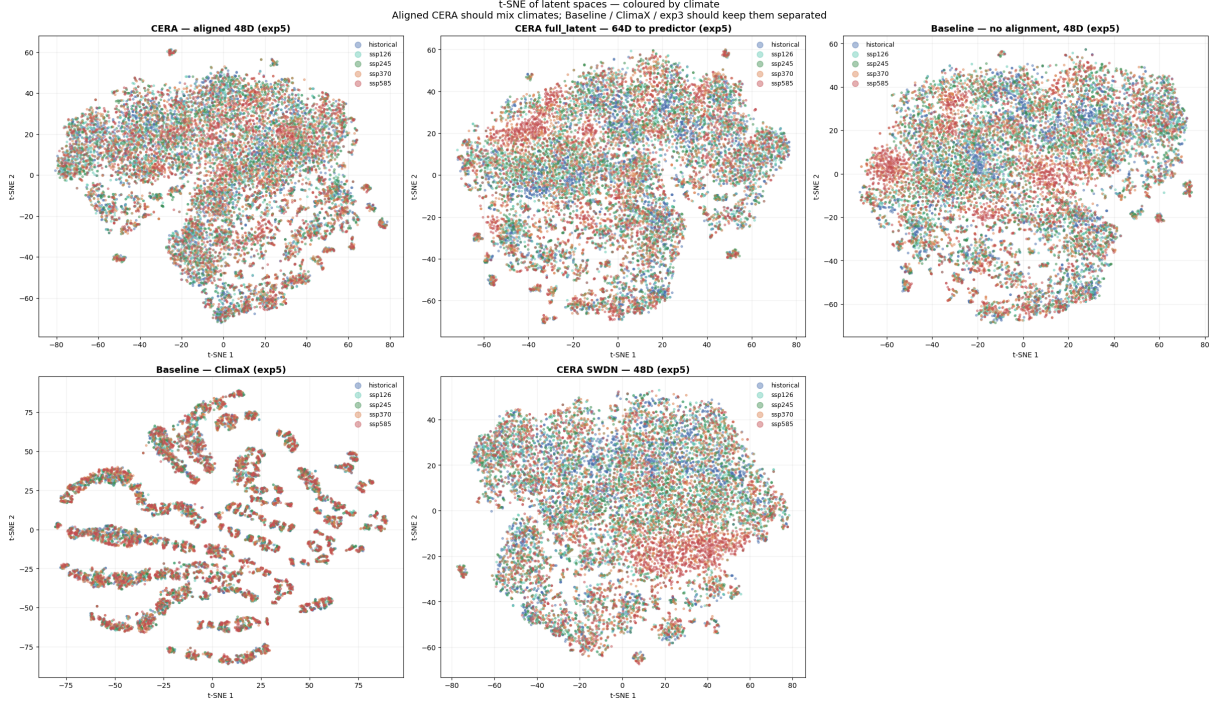


Figure 19: t-SNE projections of latent representations colored by climate scenario. Strong mixing of colors indicates weak climate separability and therefore greater climate invariance.

To quantify these observations, classifiers are trained to predict the climate scenario associated with each latent representation. We consider both a linear logistic-regression classifier and a nonlinear multilayer perceptron (MLP). Since five climates are considered, random guessing corresponds to an accuracy of 20%.

The classification results, shown in Figure 20, confirm the visual observations. The aligned representations of Exp4 and CERA consistently achieve lower classification accuracies than their non-aligned counterparts, demonstrating that explicit alignment reduces the amount of climate information accessible within the latent space. The difference is particularly visible for the nonlinear classifier, indicating that climate information is not merely hidden in a nonlinear form but is genuinely less recoverable.

The Full-Latent variant exhibits higher classification accuracy than standard CERA, suggesting that the additional non-aligned dimensions retain some climate-specific information.

As seen in the tSNE, ClimaX achieves relatively low classification accuracies, confirming that large-scale pretraining can lead to representations that are partially climate-invariant despite the absence of explicit alignment.

Taken together, these results provide direct evidence that the alignment mechanisms introduced in CERA successfully reduce climate separability while preserving useful information. Combined with the geometric analyses of the previous section, they indicate that the learned latent representations become progressively less dependent on climate scenario.

However, climate invariance alone does not guarantee physical meaning. A representation may become insensitive to climate labels while still relying on abstract statistical regularities. The next question is therefore whether the structures identified here correspond to physically meaningful quantities and relationships, which motivates the physical-invariant analyses presented in the following section.

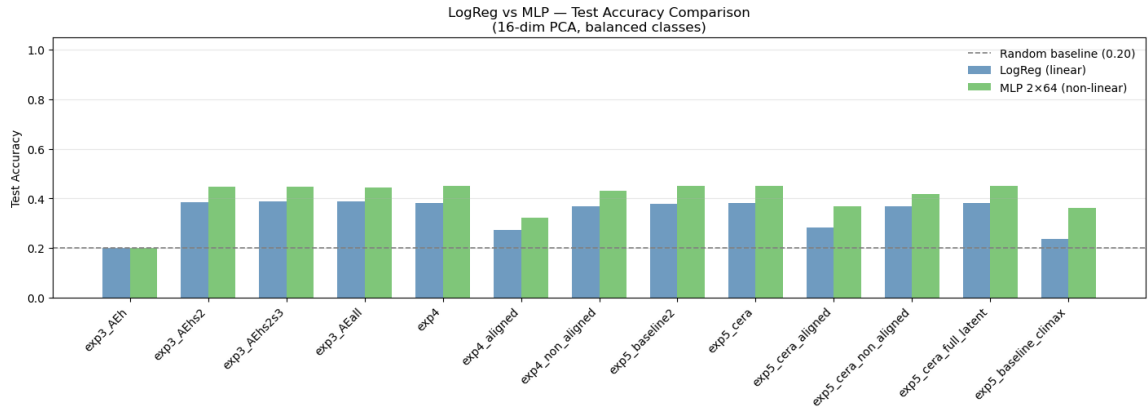


Figure 20: Climate-classification accuracy obtained from latent representations. Lower accuracy indicates stronger climate invariance. The dashed line corresponds to random guessing (20%).

7 Physical Invariant Analysis

To better understand the nature of the learned representations, we now investigate whether known physical invariants are encoded within the latent space and whether they contribute to the observed climate alignment. We focus on a set of candidate invariants motivated by atmospheric physics, including relative humidity at multiple pressure levels, vertical stability indicators, temperature gradients, wind shear, and wind magnitude.

7.1 Analysis of Known Physical Invariants

We first investigate whether the candidate physical invariants are explicitly organized within the latent spaces learned by the different models.

Figure 21 illustrates the latent representations colored according to the value of the surface-relative-humidity invariant (RH_{850}). If a latent representation has successfully encoded this quantity, neighboring points in latent space should exhibit similar colors and coherent large-scale structures should emerge.

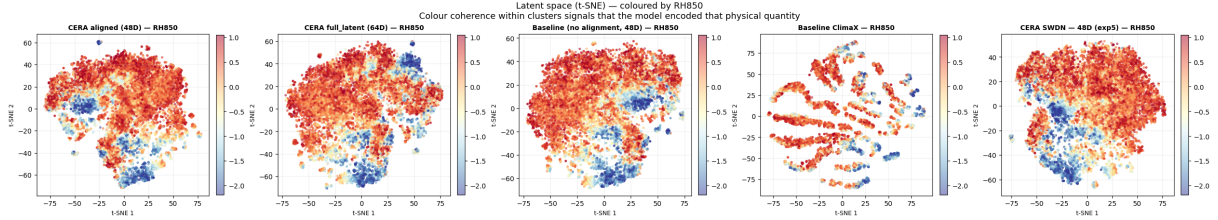


Figure 21: t-SNE projections colored by the value of the RH_{850} invariant. Spatial coherence indicates that nearby latent points share similar invariant values.

All considered models exhibit a clear organization of relative humidity within their latent representations. Large coherent regions associated with similar humidity values emerge in every latent space, indicating that this physically meaningful quantity is strongly encoded regardless of the alignment strategy. Qualitatively, the differences between methods remain relatively subtle, making it difficult to determine from visualization alone whether one representation captures the invariant more effectively than another.

To obtain a quantitative assessment, we evaluate how well invariant values can be recovered from latent neighborhoods. For each invariant, a k -nearest-neighbor regressor ($k = 10$) predicts the invariant value using neighboring latent samples. High prediction accuracy indicates that the invariant varies smoothly within the latent geometry and is therefore strongly encoded by the representation.

The results reveal that all representations encode the selected invariants remarkably well. Most R^2 values remain above 0.8, and several exceed 0.9, confirming that latent proximity is strongly related to similarity in physically meaningful quantities.

Among the compared approaches, CERA and its variants generally achieve the highest probing scores. In contrast, ClimaX displays noticeably lower probing accuracy for a number of variables, particularly RH_{500} , shear-related quantities, and lapse-rate indicators, despite exhibiting strong climate invariance in the previous section.

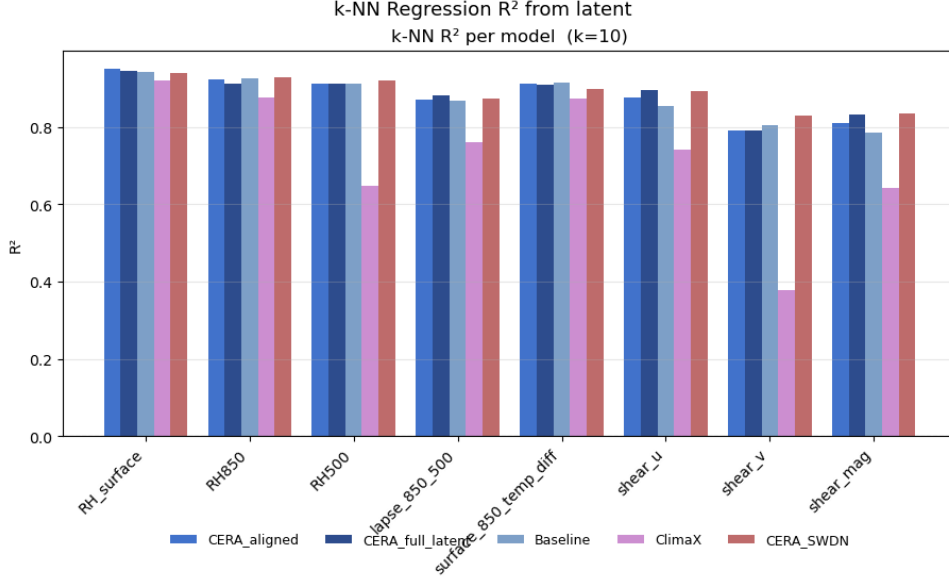


Figure 22: k -NN probing performance for selected physical invariants. Higher R^2 values indicate that latent neighborhoods preserve the corresponding physical quantity.

These results indicate that the candidate physical invariants are clearly represented within the latent geometry and remain accessible through local neighborhood relationships. However, this analysis alone does not determine whether the observed physical invariants are consistent across different climates. To address this question, we now examine pairs of neighboring samples originating from different climates and investigate whether their latent proximity corresponds to physical similarity.

7.2 Cross-Climate Pairs Study

The previous analysis showed that several physically meaningful quantities are encoded within the latent representations. We now investigate whether these quantities are also preserved across the cross-climate correspondences established by the different models.

For each historical sample, we identify neighboring samples originating from SSP climates and compare their physical properties. If latent alignment captures meaningful climate-invariant structure, neighboring points from different climates should exhibit similar values of the considered invariants.

Figure 23 summarizes the distribution of invariant differences observed across all identified climate pairs. Lower values indicate that neighboring historical and SSP samples correspond to more similar physical states.

All learned representations substantially outperform random pairing, confirming that cross-climate neighborhoods are far from arbitrary. Historical samples are generally matched with SSP samples exhibiting similar physical characteristics. Interestingly, the differences between learned representations remain relatively small. CERA, NoAlign, SWDN, and ClimaX all produce physically coherent cross-climate pairs, with median invariant differences significantly below the random baseline. This suggests that multiple representation-learning strategies are capable of identifying similar atmospheric states across climates.

ClimaX nevertheless stands apart from the other approaches, exhibiting larger invariant differences across climate pairs. This indicates that the physical quantities considered here are somewhat less well preserved across climates in its latent representation. Interestingly, this finding is consistent with the lower k -NN probing performance observed in Figure 22, suggesting that these candidate invariants are less explicitly structured within the ClimaX latent space.

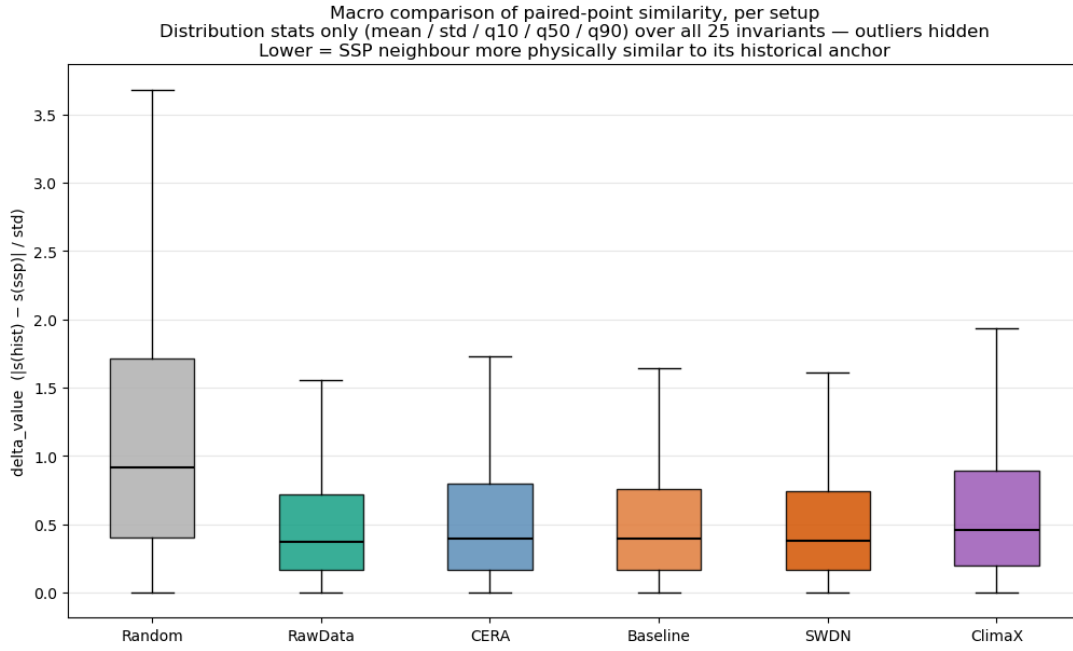


Figure 23: Distribution of invariant differences between historical samples and their nearest SSP neighbors. Lower values indicate greater physical similarity across climates.

To determine whether latent geometry itself reflects physical similarity, we additionally compute the Pearson correlation between latent-space distance and invariant differences. Positive correlations indicate that pairs located farther apart in latent space also differ more strongly in the corresponding physical quantity.

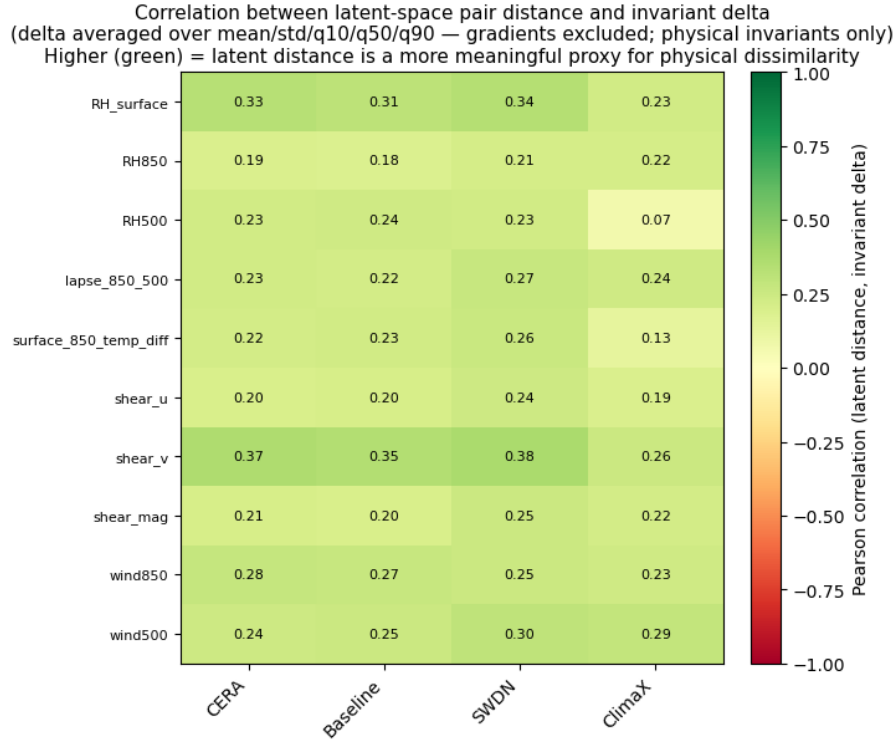


Figure 24: Correlation between latent-space distance and invariant differences across climate pairs. Higher values indicate that latent distance better reflects physical dissimilarity.

All correlations are positive, confirming that latent distance provides a meaningful proxy

for physical similarity. The strongest relationships are observed for surface relative humidity, meridional wind shear, and wind magnitude, with correlation values frequently reaching 0.3–0.4. Among the considered approaches, SWDN generally exhibits the highest correlations across most invariants, suggesting that its latent geometry more faithfully reflects physical differences between atmospheric states. CERA and the NoAlign baseline display very similar behaviors, while ClimaX tends to produce weaker correlations despite its strong predictive performance.

Taken together, these results indicate that neighboring points identified across climates correspond to physically coherent atmospheric states and that latent distances retain meaningful information about invariant similarity. However, the analysis remains restricted to a predefined set of candidate invariants. We therefore now move beyond these manually selected quantities and investigate whether additional invariant relationships can be discovered directly from the learned representations.

7.3 Physical Invariant Discovery

While the previous analyses focused on a predefined set of candidate physical invariants, it remains possible that the latent representations encode other physically meaningful relationships that were not explicitly considered. To investigate this possibility, we performed an invariant-discovery analysis using both LASSO regression and symbolic regression.

The objective is not to recover exact conservation laws, but rather to identify combinations of atmospheric variables that appear to organize the invariant directions learned in latent space. Such combinations may provide insights into the physical processes underlying the climate-invariant representations discovered by the different models.

LASSO-based variable attribution. As a first step, we applied LASSO regression to identify which variables contribute most strongly to the invariant directions extracted from each latent representation.

Unlike symbolic regression, the purpose of this analysis is not to discover explicit invariant formulas. Because the recovered coefficients depend on feature normalization and combine variables with different physical units, the resulting linear combinations cannot be interpreted directly as physical laws. Instead, LASSO serves as a feature-attribution tool that highlights the variables most strongly involved in the organization of the latent space.

For CERA, the largest coefficients are associated with humidity-related quantities (*huss*, *hus850*, *RH850*), thermodynamic variables (*ta500*), and large-scale dynamical descriptors (*zg500*, *psl*). This suggests that the aligned representation is primarily structured around a combination of thermodynamic and circulation-related information.

The SWDN variant exhibits a similar behavior, with pressure, temperature, humidity, and circulation variables appearing among the dominant contributors. In contrast, the NoAlign representation places relatively more emphasis on pressure and temperature variables and recovers fewer humidity-related descriptors. ClimaX also highlights humidity, pressure, and large-scale circulation variables, indicating that physically meaningful atmospheric quantities contribute substantially to its latent organization despite the absence of explicit alignment constraints.

The complete LASSO coefficients for all setups are provided in the Appendix.

Symbolic regression. To move beyond variable attribution and search for explicit candidate invariants, we applied symbolic regression to the latent invariant directions. The objective of this analysis is to identify simple analytical expressions that remain approximately constant across neighboring cross-climate pairs.

Table 3 reports the ten highest-ranked formulas discovered for the CERA representation.

Table 3: Top symbolic-regression formulas discovered for CERA.

Formula
$psl - huss$
$tas - huss$
$ta850 - ua500$
$tas + ua850 - sfcWind$
$ua850 + huss$
$huss + sfcWind$
$ta850 - hus850$
$0.629 + wap500 - tas + va500$
$huss - ua850$
$wap500 - ua850$

Several observations can be made from these results. First, the recovered expressions are generally sparse and involve only a small number of atmospheric variables. Humidity variables ($huss$, $hus850$), temperature variables (tas , $ta850$), circulation descriptors ($ua850$, $ua500$, $va500$), and vertical motion ($wap500$) appear repeatedly among the highest-ranked candidates. This is broadly consistent with the conclusions drawn from the LASSO analysis and suggests that these quantities play an important role in the organization of the latent space.

However, despite their apparent simplicity, the recovered formulas do not correspond to any known physical invariant or conservation law. While some expressions resemble physically meaningful contrasts between thermodynamic and dynamical quantities, none can be directly interpreted as a recognized atmospheric quantity such as relative humidity, moist static energy, atmospheric stability, or wind shear.

Consequently, the symbolic-regression analysis does not provide evidence that CERA explicitly learns identifiable physical invariants. Instead, it suggests that the latent invariant directions are organized around combinations of humidity, temperature, pressure, and circulation variables that remain difficult to interpret physically. These expressions should therefore be viewed as statistical descriptors of the learned representation rather than as newly discovered physical laws.

Taken together, the LASSO and symbolic-regression analyses indicate that the learned climate-invariant representations are clearly influenced by physically relevant atmospheric variables. Nevertheless, the present methodology was not able to recover explicit physical invariants from the latent space. While the representations appear to encode meaningful physical structure, the precise nature of this structure remains difficult to characterize and warrants further investigation.

8 Discussion

The objective of this internship was to investigate the relationship between latent representation alignment, climate invariance, and out-of-distribution climate generalization. More specifically, we sought to answer three main questions: whether explicit climate alignment improves generalization under climate change, whether the resulting latent representations encode meaningful climate-invariant structures, and how such approaches compare to large pretrained climate foundation models such as ClimaX.

Overall, the results provide a nuanced picture of climate-invariant representation learning. While explicit alignment successfully modifies the geometry of latent spaces and promotes climate invariance, its impact on downstream predictive performance appears more limited than initially expected. At the same time, the analyses reveal that physically meaningful structures are consistently encoded within the learned representations, although the precise nature of these structures remains difficult to characterize.

Does climate alignment improve climate generalization?

The climate generalization experiments indicate that the answer depends strongly on the prediction task considered.

For temperature prediction, all approaches generalize relatively well across climates. Differences between CERA and its non-aligned counterpart remain small, suggesting that explicit climate invariance is not essential for this task. Temperature is a relatively smooth variable whose governing relationships remain stable across climates, making it less sensitive to distribution shifts.

The situation is markedly different for precipitation prediction. As expected, precipitation constitutes a substantially more challenging out-of-distribution problem, and all models experience some degradation as warming increases. Climate-aligned approaches consistently outperform the simple supervised baseline and remain competitive under severe warming scenarios. However, the difference between CERA and the NoAlign baseline remains surprisingly modest. This result suggests that a significant fraction of the robustness originates from the representation-learning framework itself rather than from the alignment objective alone.

One of the most striking findings is the strong performance of ClimaX. Despite exhibiting weaker physical organization according to several latent-space analyses, ClimaX achieves the best overall precipitation prediction accuracy and remains highly robust under climate change. This observation highlights the effectiveness of large-scale pretraining as an alternative strategy for improving climate generalization.

Taken together, these results suggest that explicit climate alignment can contribute to climate robustness, but its benefits may be smaller than originally anticipated when compared to the gains provided by representation learning and large-scale pretraining.

Does CERA learn climate-invariant representations?

The latent-space analyses provide a much clearer answer to this question.

Across all geometric metrics considered, explicit alignment consistently reduces the separation between climate distributions. CERA produces stronger climate mixing in t-SNE visualizations, smaller cross-climate distances, and lower climate-classification accuracies than comparable non-aligned architectures. These observations demonstrate that the alignment objective successfully removes climate-specific information from the latent representation.

However, the analyses also reveal that alignment is not achieved solely through a reorganization of latent geometry. Standard SWD alignment induces a substantial contraction of the

latent space, suggesting that part of the alignment effect originates from global geometric rescaling. This observation motivated the introduction of the SWDN variant, which attempts to disentangle genuine climate alignment from simple latent-space compression.

Interestingly, ClimaX also exhibits relatively strong climate invariance despite the absence of any explicit alignment objective. This suggests that climate-invariant structures may naturally emerge from large-scale pretraining on diverse climate datasets, although through mechanisms that differ from those employed by CERA.

Overall, the results provide strong evidence that CERA does learn representations that are more climate-invariant than those learned without alignment. The remaining question is therefore whether these invariant representations correspond to meaningful physical structures.

Do the learned representations correspond to physical invariants?

This question proved to be the most challenging and arguably the most interesting aspect of the project.

The analyses based on known physical invariants show that latent representations are generally organized around physically meaningful quantities such as relative humidity, atmospheric stability, temperature gradients, and wind-related variables. Neighboring samples in latent space tend to correspond to similar physical states, both within and across climates, and latent distances correlate positively with differences in these quantities. However, the strength of this organization varies across models. CERA, SWDN, and the NoAlign baseline consistently achieve the highest probing scores and strongest correlations, suggesting that these representations capture physical invariant structure particularly well. ClimaX also encodes substantial physical information, but its probing performance and cross-climate invariant consistency are systematically lower, indicating that physical invariants are somewhat less explicitly organized in its latent space. Nevertheless, its performance remains well above random baselines, showing that physically relevant information is still present within the representation.

These findings suggest that climate alignment does not simply remove climate labels from the representation. Instead, it promotes an organization that remains closely connected to meaningful atmospheric properties.

Nevertheless, the invariant-discovery experiments reveal the limits of the current analysis. Neither LASSO nor symbolic regression succeeded in recovering explicit physical invariants or recognizable conservation laws. While the discovered formulas consistently involve physically relevant variables such as humidity, temperature, pressure, and circulation descriptors, they remain difficult to interpret in terms of established atmospheric theory.

Consequently, the present work provides evidence that physically meaningful information is encoded within the learned representations, but it does not demonstrate the emergence of identifiable physical invariants. The latent spaces appear to capture robust physical structure, yet the precise nature of this structure remains largely opaque.

Implications for climate foundation models

One of the motivations of this study was to understand how explicit climate-invariant learning compares with foundation-model approaches.

The results suggest that these strategies should not necessarily be viewed as competing alternatives. ClimaX achieves excellent predictive performance and strong climate robustness, demonstrating the power of large-scale pretraining. At the same time, several analyses indicate that its latent representations are somewhat less organized around the candidate physical invariants considered in this work than those learned by CERA.

This observation raises an intriguing possibility. Rather than replacing foundation models, explicit climate-alignment mechanisms may provide a complementary inductive bias that encourages physically meaningful organization within large pretrained representations. Future climate foundation models could potentially benefit from combining both approaches: large-scale pre-training to learn rich representations and explicit invariance objectives to improve robustness under climate change.

Limitations and future work

Several limitations of the present study suggest promising directions for future research.

First, the alignment procedure was restricted to a single auxiliary climate (SSP2-4.5). Extending the framework to simultaneously align multiple future climates could encourage the emergence of more robust invariant structures and better reflect the diversity of future climate trajectories.

Second, the present work focuses exclusively on a single CMIP6 model (CESM2). Evaluating climate-invariant learning across multiple Earth System Models would provide a stronger test of generalization and would allow climate invariance to be studied across both climate scenarios and model structures.

Third, while the latent-space analyses provide valuable insights into representation geometry, the interpretation of the learned invariants remains largely unresolved. More sophisticated methods from explainable machine learning, causal representation learning, or mechanistic interpretability may provide a deeper understanding of the structures encoded by climate-aligned representations.

Finally, the strong performance of ClimaX suggests that future work should explore hybrid approaches combining climate foundation models with explicit alignment objectives. Such methods may offer a promising path toward representations that are simultaneously accurate, robust, and physically interpretable.

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A Model Architectures and Hyperparameters

This appendix provides implementation details for the architectures and training procedures used throughout this study. Only parameters that were fixed explicitly in the experimental notebooks are reported. Unless otherwise stated, all models were implemented in PyTorch and trained using the same preprocessed CMIP6 samples described in Section 3.

A.1 Dataset and Experimental Configuration

Each sample consists of a multivariate atmospheric patch containing 70 grid points arranged on a 10×7 spatial grid. For a given prediction task, one of the sixteen climate variables is used as the target and removed from the inputs, leaving fifteen input channels. The resulting input dimension is therefore

$$15 \times 70 = 1050.$$

The primary experiments use up to 10^6 samples per climate. The samples are divided independently within each climate into training, validation, and test sets using the proportions reported in Table 4. The same split indices are used when comparing models whenever possible.

Table 4: Main data and experimental parameters.

Parameter	Value
Spatial patch shape	10×7 grid points
Grid points per patch	70
Input variables per task	15
Input dimension	1050
Maximum absolute latitude	30°
Approximate patch size	$1000 \text{ km} \times 1000 \text{ km}$
Temporal resolution	Daily
Maximum samples per climate	10^6
Validation fraction	0.05
Test fraction	0.15
Random split seed	42

All variables are standardized using statistics computed exclusively from the Historical training set. The same normalization parameters are subsequently applied to all climate scenarios. Precipitation is first converted to mm day^{-1} and transformed according to

$$pr_{\log} = \log(1 + pr)$$

before standardization.

A.2 Convolutional Autoencoder

The autoencoder used in Exp3, Exp4, CERA, NoAlign, and the CERA variants is based on a two-dimensional convolutional architecture. The flattened input is first reshaped into a tensor of shape

$$(B, 15, 10, 7),$$

where B denotes the batch size.

The encoder consists of the following sequence:

$$15 \xrightarrow{\text{Conv2D}(3 \times 3)} 32 \xrightarrow{\text{ReLU}} 32 \xrightarrow{\text{Conv2D}(3 \times 3)} 32 \xrightarrow{\text{ReLU}} \xrightarrow{\text{Conv2D}(3 \times 3, \text{stride}=2)} 64 \xrightarrow{\text{ReLU}} \xrightarrow{\text{Conv2D}(3 \times 3)} 128 \xrightarrow{\text{ReLU}} .$$

The resulting feature map is flattened and projected through a fully connected layer to a 64-dimensional latent representation:

$$z \in \mathbb{R}^{64}.$$

The decoder first projects the latent vector back to the encoded feature-map shape. It then applies a convolutional refinement block composed of:

$$128 \rightarrow 64 \rightarrow 32 \rightarrow 32 \rightarrow 15,$$

with ReLU activations between layers. Bilinear upsampling restores the original 10×7 spatial resolution before the final convolution.

The reconstruction objective is computed over all 64 latent dimensions. In aligned models, the latent representation is partitioned into

$$z = (z_{\text{align}}, z_{\text{free}}),$$

with

$$z_{\text{align}} \in \mathbb{R}^{48}, \quad z_{\text{free}} \in \mathbb{R}^{16}.$$

Only the first 48 dimensions are subjected to the alignment objective. Except in the CERA-Full variant, these are also the only dimensions provided to the downstream predictor.

A.3 Alternative MLP Autoencoder

An MLP autoencoder was also implemented as an alternative to the convolutional architecture. Its encoder has the form

$$1050 \rightarrow h_1 \rightarrow h_2 \rightarrow 64,$$

and its decoder mirrors this structure:

$$64 \rightarrow h_2 \rightarrow h_1 \rightarrow 1050.$$

The hidden dimensions are determined from the input dimension according to

$$h_1 = \max \left(256, \min \left(1024, \frac{d_{\text{in}}}{2} \right) \right),$$

$$h_2 = \max \left(128, \min \left(512, \frac{d_{\text{in}}}{8} \right) \right).$$

For $d_{\text{in}} = 1050$, this gives

$$h_1 = 525, \quad h_2 = 131.$$

ReLU activations are used after each hidden layer. The main experiments reported in the body of the report use the CNN autoencoder in order to preserve the two-dimensional spatial organization of the climate patches.

A.4 Prediction Network

The predictor used by CERA, NoAlign, CERA-Full, and the Simple and Physical baselines is a multilayer perceptron composed of five hidden layers. Each hidden layer contains 128 units and uses a LeakyReLU activation with negative slope 0.1:

$$d_{\text{in}} \rightarrow 128 \rightarrow 128 \rightarrow 128 \rightarrow 128 \rightarrow 128 \rightarrow d_{\text{out}}.$$

For CERA and NoAlign,

$$d_{\text{in}} = 48,$$

whereas CERA-Full receives the complete latent vector:

$$d_{\text{in}} = 64.$$

For the Simple baseline, the predictor receives the flattened normalized climate patch directly. The Physical baseline uses the same predictor architecture but augments the raw input with manually constructed physical and statistical features.

The output dimension is 70 for the temperature and precipitation tasks, corresponding to one prediction per grid point in the patch. The same general prediction architecture is adapted to the masked-reconstruction target.

A.5 Climate Alignment Objectives

A.5.1 Sliced Wasserstein Distance

The standard CERA implementation uses the Sliced Wasserstein Distance as a computational approximation of the Earth Mover’s Distance used in the original CERA study. Let

$$Z_h = \{z_i^h\}_{i=1}^n \quad \text{and} \quad Z_s = \{z_i^s\}_{i=1}^n$$

denote batches of aligned latent representations from the Historical and SSP2-4.5 climates.

For each randomly sampled unit direction θ_k , the latent vectors are projected onto a one-dimensional axis:

$$p_i^{h,k} = \theta_k^\top z_i^h, \quad p_i^{s,k} = \theta_k^\top z_i^s.$$

After sorting the projected values, the alignment loss is

$$\mathcal{L}_{\text{SWD}} = \frac{1}{K} \sum_{k=1}^K \frac{1}{n} \sum_{i=1}^n \left| p_{(i)}^{h,k} - p_{(i)}^{s,k} \right|,$$

where $K = 64$ random projections are used during training.

A.5.2 Normalized Sliced Wasserstein Distance

The SWDN variant normalizes the latent representations before computing the Sliced Wasserstein Distance. This modification reduces the sensitivity of the objective to global contractions or expansions of the latent space. Its purpose is to prevent the model from reducing the alignment loss mainly through geometric rescaling rather than through a modification of relative latent organization.

A.5.3 Adversarial Alignment

The adversarial variant replaces distribution matching with a domain-confusion objective. A domain classifier receives the aligned latent representation and predicts whether it originates from the Historical or SSP2-4.5 climate.

The classifier architecture is

$$48 \rightarrow 128 \rightarrow 64 \rightarrow 2,$$

with ReLU activations. A gradient-reversal layer is placed between the encoder and domain classifier. The classifier is optimized to identify the climate correctly, while the reversed gradient encourages the encoder to remove climate-discriminative information.

A.6 Composite CERA Objective

The CERA-like architectures are optimized using the same three-term objective as the original framework:

$$\mathcal{L} = (1 - \lambda_{\text{rec}} - \lambda_{\text{align}})\mathcal{L}_{\text{pred}} + \lambda_{\text{rec}}\mathcal{L}_{\text{rec}} + \lambda_{\text{align}}\mathcal{L}_{\text{align}}.$$

In the implementation, the relative weights are parameterized through the prediction and alignment coefficients used in the training notebooks. The main values used throughout the experiments are

$$\lambda_{\text{align}} = 10^{-4}, \quad \lambda_{\text{pred}} = 10^{-2}.$$

The prediction loss is computed only on labeled Historical samples. Reconstruction is evaluated on both Historical and SSP2-4.5 inputs, while the alignment loss compares their first 48 latent dimensions.

A.7 Training Hyperparameters

Table 5 summarizes the main hyperparameters used by the autoencoder and CERA-like architectures.

Table 5: Training hyperparameters for the autoencoder and CERA-like experiments.

Parameter	Exp3	Exp4	CERA-like models
Autoencoder type	CNN	CNN	CNN
Latent dimension	64	64	64
Aligned dimensions	–	48	48
Epoch limit	100	100	100
Batch size	128	256	256
Learning rate	10^{-3}	10^{-3}	10^{-3}
Weight decay	10^{-3}	10^{-3}	10^{-3}
Optimizer	Adam	AdamW	AdamW
Early-stopping patience	5	–	10
SWD projections	–	64	64
Alignment coefficient	–	10^{-4}	10^{-4}
Prediction coefficient	–	–	10^{-2}
Evaluation batch size	–	–	2048

For SWDN experiments, a larger batch size of 1024 is used to obtain more stable batch-level normalization statistics.

The Simple and Physical baselines use the same five-layer predictor as CERA. Their main training parameters are provided in Table 6.

Table 6: Training hyperparameters for the supervised MLP baselines.

Parameter	Value
Epoch limit	100
Batch size	256
Learning rate	10^{-3}
Weight decay	10^{-3}
Optimizer	AdamW
Hidden layers	5
Hidden width	128
Activation	LeakyReLU (0.1)
Early-stopping patience	10
Training climate	Historical

A.8 ClimaX Baseline

The ClimaX baseline uses a pretrained Transformer backbone with an embedding dimension of 1024. Climate patches are tokenized using variable-specific 2×2 convolutional patch embeddings and processed by the pretrained Transformer blocks.

The pretrained backbone is frozen during the downstream experiments. Its output features are computed once and subsequently used to train a prediction head. A progressive compression module maps the 1024-dimensional pretrained features to a 64-dimensional representation, allowing comparison with the CERA latent space.

The ClimaX prediction head uses five hidden layers of width 128. Its main hyperparameters are reported in Table 7.

Table 7: Main hyperparameters of the ClimaX baseline.

Parameter	Value
Pretrained embedding dimension	1024
Extracted latent dimension	64
Hidden layers in prediction head	5
Hidden width	128
Epoch limit	100
Training batch size	512
Backbone precomputation batch size	256
Evaluation batch size	512
Learning rate	3×10^{-4}
Weight decay	10^{-5}
Optimizer	AdamW
Early-stopping patience	15
Fine-tuning climate	Historical
Backbone optimization	Frozen

A.9 Manually Constructed Physical Features

The Physical baseline augments the original climate variables with ten manually constructed physical quantities.

Relative humidity is computed at the surface, 850 hPa, and 500 hPa. Given specific humidity q , temperature T , and pressure p , vapor pressure is estimated as

$$e = \frac{qp}{\epsilon + (1 - \epsilon)q}, \quad \epsilon = 0.622,$$

while saturation vapor pressure is approximated by

$$e_s = 611.2 \exp \left[\frac{17.67(T - 273.15)}{T - 29.65} \right].$$

Relative humidity is then

$$RH = \frac{e}{e_s}.$$

The complete set of physical features is reported in Table 8.

Table 8: Physically motivated features used by the Physical baseline and in the latent-space analysis.

Feature	Definition
RH_{surface}	Relative humidity computed from <code>huss</code> , <code>tas</code> , and <code>psl</code>
RH_{850}	Relative humidity computed from <code>hus850</code> and <code>ta850</code>
RH_{500}	Relative humidity computed from <code>hus500</code> and <code>ta500</code>
<code>lapse_850_500</code>	<code>ta850</code> – <code>ta500</code>
<code>surface_850_temp_diff</code>	<code>tas</code> – <code>ta850</code>
<code>shear_u</code>	<code>ua500</code> – <code>ua850</code>
<code>shear_v</code>	<code>va500</code> – <code>va850</code>
<code>shear_mag</code>	$\sqrt{(\text{ua500} - \text{ua850})^2 + (\text{va500} - \text{va850})^2}$
<code>wind850</code>	$\sqrt{\text{ua850}^2 + \text{va850}^2}$
<code>wind500</code>	$\sqrt{\text{ua500}^2 + \text{va500}^2}$

The Physical baseline additionally includes one statistical anomaly for each original input variable:

$$X'(t, x, y) = X(t, x, y) - \bar{X}_{m(t)}^{\text{hist, train}}(x, y),$$

where $\bar{X}_{m(t)}^{\text{hist, train}}$ is the Historical training climatology for the corresponding calendar month and spatial location.

A.10 Latent-Space Metrics

A.10.1 Centroid Distance

For a latent distribution Z_c associated with climate c , its centroid is

$$\mu_c = \frac{1}{N_c} \sum_{i=1}^{N_c} z_i.$$

The centroid distance between the Historical climate and climate c is

$$d_{\text{centroid}}(c) = \|\mu_c - \mu_{\text{hist}}\|_2.$$

A.10.2 RMS Dispersion

The RMS dispersion of a latent distribution is computed as

$$\text{RMS}(Z_c) = \sqrt{\frac{1}{N_c} \sum_{i=1}^{N_c} \|z_i - \mu_c\|_2^2}.$$

This metric characterizes the global spatial scale of the latent representation and is used to detect geometric contraction induced by alignment.

A.10.3 Effective Rank

Let

$$\lambda_1, \dots, \lambda_d$$

denote the eigenvalues of the latent covariance matrix. The normalized spectrum is

$$p_i = \frac{\lambda_i}{\sum_{j=1}^d \lambda_j}.$$

The effective rank is defined as

$$r_{\text{eff}} = \exp \left(- \sum_{i=1}^d p_i \log p_i \right).$$

Because the compared latent spaces do not always have the same dimensionality, effective-rank values are interpreted relative to their maximum available dimension.

A.11 Climate Classification Protocol

Climate separability is assessed by predicting the source climate from the learned representation. The five considered classes are Historical, SSP1-2.6, SSP2-4.5, SSP3-7.0, and SSP5-8.5. Classes are balanced before training, giving a random baseline accuracy of

$$\frac{1}{5} = 0.20.$$

Before classification, latent representations are projected onto 16 PCA components. Two classifiers are considered:

- a multinomial logistic-regression classifier, used to measure linearly accessible climate information;
- a nonlinear MLP with two hidden layers of width 64, used to detect climate information encoded through nonlinear relationships.

B Complementary Results

B.1 Detailed Reconstruction Results

Table 9 reports the reconstruction R^2 obtained for each atmospheric variable and each autoencoder-based setup. Presenting the scores variable by variable makes it possible to verify that the high average reconstruction performance is not driven by only a small subset of easily reconstructed fields.

With the exception of AEh, reconstruction remains consistently high across nearly all variables and architectures. The lowest scores are generally obtained for **rsds** and **wap500**, which appear more difficult to reconstruct than pressure, temperature, humidity, wind, and geopotential-height fields. Nevertheless, these scores remain close to or above $R^2 = 0.88$ for all non-collapsed setups.

AEh behaves very differently. Its reconstruction scores are close to zero or negative for most variables, with particularly poor values for **ta500** and **zg500**. This confirms that its low average score is not caused by a single problematic variable but reflects a broadly degenerate representation. Results derived from the AEh latent space should therefore be interpreted with caution. By contrast, all other representations preserve substantial information across the full range of atmospheric variables, supporting the relevance of the subsequent geometric and physical analyses.

Table 9: Variable-level reconstruction R^2 for the Exp3 and Exp4 representations.

Variable	AEh	AEhs2	AEhs2s3	AEall	Exp4
hus500	-0.041	0.961	0.963	0.964	0.978
hus850	-0.062	0.955	0.949	0.960	0.976
huss	-0.110	0.956	0.962	0.963	0.984
psl	-0.002	0.974	0.977	0.977	0.993
rsds	0.000	0.902	0.898	0.902	0.922
sfcWind	0.000	0.934	0.941	0.943	0.972
ta500	-0.505	0.977	0.975	0.977	0.987
ta850	-0.185	0.962	0.968	0.970	0.987
tas	-0.167	0.950	0.951	0.947	0.983
ua500	0.000	0.971	0.968	0.974	0.988
ua850	0.000	0.963	0.966	0.968	0.980
va500	0.000	0.961	0.961	0.965	0.980
va850	0.000	0.953	0.957	0.953	0.977
wap500	0.000	0.902	0.899	0.891	0.951
zg500	-0.961	0.980	0.980	0.984	0.994

Variable	CERA	NoAlign	CERA-Full	SWDN	Adversarial
hus500	0.961	0.964	0.978	0.977	0.978
hus850	0.958	0.958	0.976	0.974	0.976
huss	0.979	0.978	0.984	0.983	0.984
psl	0.982	0.986	0.992	0.993	0.993
rsds	0.879	0.877	0.921	0.916	0.921
sfcWind	0.943	0.958	0.972	0.970	0.972
ta500	0.978	0.979	0.988	0.987	0.988
ta850	0.977	0.980	0.987	0.986	0.986
tas	0.972	0.975	0.984	0.981	0.984
ua500	0.973	0.977	0.987	0.986	0.988
ua850	0.961	0.968	0.980	0.979	0.981
va500	0.961	0.967	0.979	0.979	0.981
va850	0.953	0.954	0.976	0.975	0.977
wap500	0.920	0.928	0.954	0.953	0.954
zg500	0.986	0.987	0.994	0.994	0.994

B.2 Additional Latent-Space Visualizations

The main body of the report presents representative t-SNE projections colored by climate and by RH_{850} . Additional visualizations may be included here for the remaining candidate physical quantities:

- surface relative humidity;
- relative humidity at 500 hPa;
- vertical temperature contrasts;
- zonal and meridional wind shear;
- wind magnitudes at 850 and 500 hPa.

These complementary projections are intended as qualitative diagnostics only. Quantitative comparisons are provided by the probing and cross-climate neighborhood analyses in Section 7.

B.3 Detailed Invariant-Probing Results

The probing experiments evaluate whether candidate physical quantities can be recovered from local latent neighborhoods. A k -nearest-neighbor regressor with

$$k = 10$$

is fitted for each invariant and latent representation.

The invariants evaluated in the main analysis are:

$$RH_{\text{surface}}, \quad RH_{850}, \quad RH_{500}, \quad \text{lapse}_{850-500}, \quad \Delta T_{\text{surface}-850}, \\ \text{shear}_u, \quad \text{shear}_v, \quad \text{shear}_{\text{mag}}, \quad \text{wind}_{850}, \quad \text{wind}_{500}.$$

The full probing results may be presented as an expanded version of Figure 22, including both linear and k -NN probing scores for every candidate invariant.

B.4 Cross-Climate Pair Statistics

For each Historical sample, nearest neighbors are identified in each SSP latent distribution. Physical similarity is assessed by comparing patch-level statistics of the candidate invariants. The statistics considered include

$$\text{mean}, \quad \text{standard deviation}, \quad q_{10}, \quad q_{50}, \quad q_{90},$$

as well as spatial-gradient summaries when applicable.

For an invariant I and a cross-climate pair (i, j) , the difference associated with statistic s is

$$\Delta I_s(i, j) = |s(I_i) - s(I_j)|.$$

The aggregated comparison shown in the main report combines these differences across candidate invariants and summary statistics. Lower values indicate that SSP neighbors are physically more similar to their Historical anchors.

B.5 LASSO Variable Attribution

LASSO regression is used as an exploratory variable-attribution method rather than as a direct physical-law discovery method. The optimization identifies a sparse combination of input-derived features that approximates the invariant direction extracted from each latent representation.

Because the coefficients depend on feature standardization and combine quantities with different physical dimensions, the resulting expressions are not interpreted as physical invariants. Instead, they indicate which variables and families of variables contribute most strongly to the latent organization.

For CERA, the dominant coefficients involve humidity-related variables, including **huss**, **hus850**, and RH_{850} , together with thermodynamic and dynamical variables such as **ta500**, **psl**, and **zg500**. SWDN presents a similar combination of pressure, temperature, humidity, and circulation descriptors. The NoAlign representation assigns comparatively more importance to pressure, temperature, and wind variables, whereas ClimaX emphasizes humidity, pressure, geopotential height, and mid-tropospheric temperature.

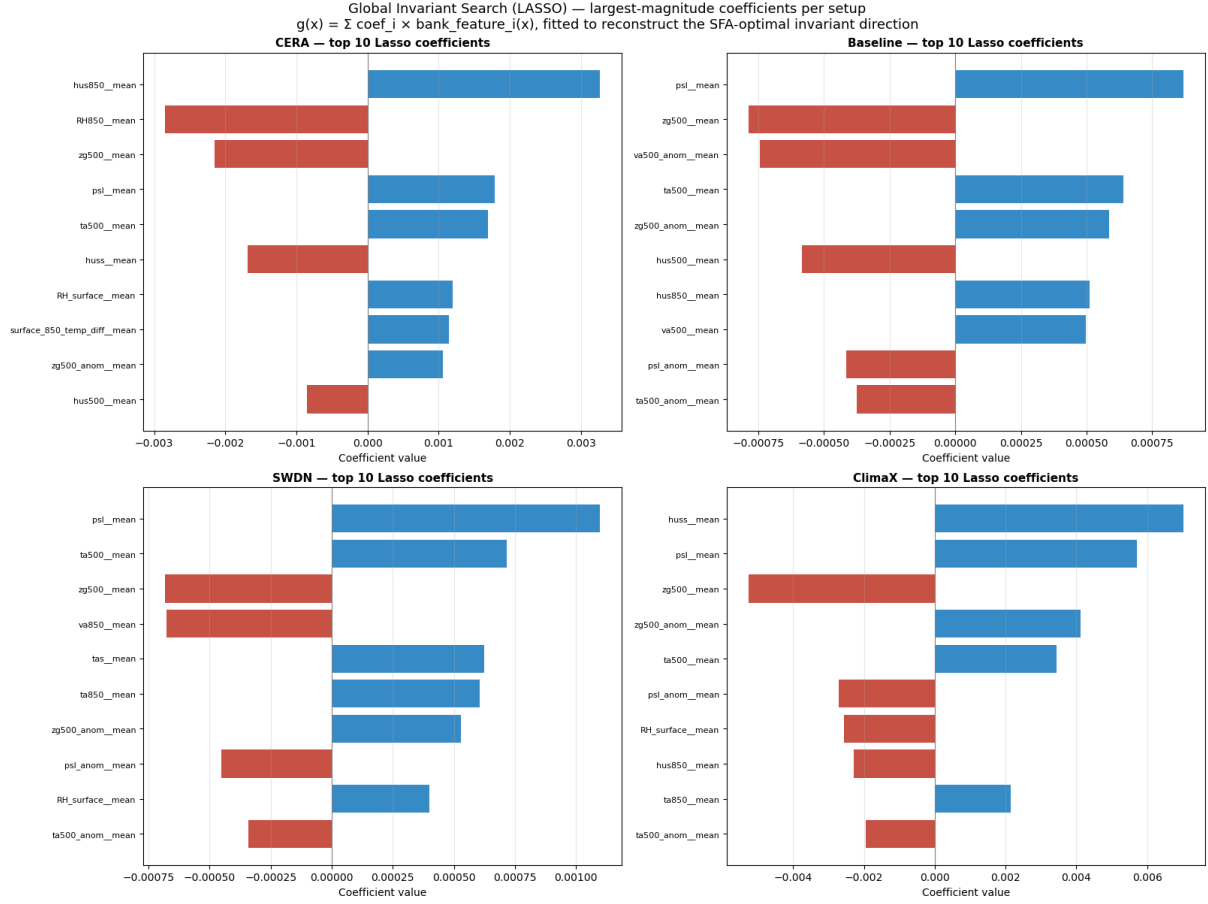


Figure 25: Largest-magnitude LASSO coefficients obtained for CERA, NoAlign, SWDN, and ClimaX. The coefficients are interpreted as variable-attribution indicators rather than explicit physical invariants.

B.6 Symbolic-Regression Candidates

Symbolic regression searches for compact analytical expressions that vary weakly across neighboring cross-climate pairs. The resulting expressions are statistical candidates and should not be interpreted as physical laws without additional theoretical and empirical validation.

B.6.1 CERA

$$g_1 = psl - huss, \quad (2)$$

$$g_2 = tas - huss, \quad (3)$$

$$g_3 = ta850 - ua500, \quad (4)$$

$$g_4 = tas + ua850 - sfcWind, \quad (5)$$

$$g_5 = ua850 + huss, \quad (6)$$

$$g_6 = huss + sfcWind, \quad (7)$$

$$g_7 = ta850 - hus850, \quad (8)$$

$$g_8 = 0.629 + wap500 - tas + va500, \quad (9)$$

$$g_9 = huss - ua850, \quad (10)$$

$$g_{10} = wap500 - ua850. \quad (11)$$

B.6.2 NoAlign

$$g_1 = 0.958 \, huss, \quad (12)$$

$$g_2 = 1 + ta850 + hus850, \quad (13)$$

$$g_3 = hus850 + hus500, \quad (14)$$

$$g_4 = tas - huss, \quad (15)$$

$$g_5 = wap500 - huss, \quad (16)$$

$$g_6 = huss + sfcWind, \quad (17)$$

$$g_7 = psl + sfcWind, \quad (18)$$

$$g_8 = hus850 - sfcWind, \quad (19)$$

$$g_9 = tas \cdot psl + tas, \quad (20)$$

$$g_{10} = wap500 - tas. \quad (21)$$

B.6.3 SWDN

$$g_1 = psl - tas - hus500 + ua850 - huss, \quad (22)$$

$$g_2 = ta850 + tas, \quad (23)$$

$$g_3 = hus850 + hus500, \quad (24)$$

$$g_4 = ta500 - psl, \quad (25)$$

$$g_5 = wap500 - huss, \quad (26)$$

$$g_6 = tas \cdot ua500 - psl + hus850, \quad (27)$$

$$g_7 = tas - huss, \quad (28)$$

$$g_8 = ua850 + huss, \quad (29)$$

$$g_9 = ua500 - hus500, \quad (30)$$

$$g_{10} = ua500 + rsds - huss. \quad (31)$$

B.6.4 ClimaX

$$g_1 = psl - huss - ta500 - hus850, \quad (32)$$

$$g_2 = tas - huss, \quad (33)$$

$$g_3 = hus850 - sfcWind, \quad (34)$$

$$g_4 = hus850 + ua850, \quad (35)$$

$$g_5 = ta850 - tas, \quad (36)$$

$$g_6 = huss + va850, \quad (37)$$

$$g_7 = huss + sfcWind, \quad (38)$$

$$g_8 = hus850 - ua850, \quad (39)$$

$$g_9 = ta850 - hus850, \quad (40)$$

$$g_{10} = ta850 \cdot psl + ta500. \quad (41)$$

Several variables recur across the four setups, particularly humidity, temperature, pressure, and wind-related quantities. However, none of the reported expressions corresponds directly to a recognized atmospheric invariant or conservation law. The formulas should therefore be interpreted as compact statistical descriptions of latent organization rather than as discovered physical invariants.